
CAPSTONE PROJECT 1

Mental Health Survey Analysis Project

*Mental Health Survey Analysis Using PostgreSQL, Python, and Power BI
An End-to-End Data Analytics and Database Management Project Report*

INSTITUTE

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Dataset: Public Google Forms Mental Health Survey

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1. Introduction

Mental health has become a significant public health concern, affecting individuals' academic performance, workplace productivity, social relationships, and overall quality of life. Factors such as stress, anxiety, burnout, poor sleep quality, work pressure, and limited social support increasingly influence the wellbeing of students and working professionals. Understanding these factors through data-driven approaches can help identify patterns, support early intervention strategies, and inform wellbeing initiatives.

This project presents an end-to-end data analytics and database management solution for analyzing mental health survey data collected through a custom-designed Google Form questionnaire. The survey gathered responses related to demographic characteristics, lifestyle habits, and mental health indicators from participants across various states in India. After collecting the responses, the data was exported as an Excel dataset and underwent cleaning, validation, and preprocessing to ensure consistency and analytical readiness.

To establish a structured and scalable data architecture, the dataset was transformed into a normalized relational database design consisting of three interconnected entities: Respondent, Lifestyle_Habits, and Mental_Health_Assessment. One-to-Many (1:N) relationships were established using primary and foreign key constraints to maintain referential integrity. The database schema was implemented in PostgreSQL, where the cleaned survey data was imported and validated through SQL operations.

Following database implementation, Python was utilized for data preparation, exploratory data analysis (EDA), feature engineering, and statistical visualization. Various analytical techniques were employed to investigate relationships between stress levels, burnout, anxiety, depression, sleep quality, work pressure, physical activity, screen time, coping mechanisms, and social support factors.

Finally, Power BI dashboards were developed to transform the analytical findings into interactive and user-friendly visualizations. The dashboards provide dynamic exploration capabilities through filters and slicers, enabling stakeholders to examine mental health trends across different demographic groups and lifestyle categories.

By integrating survey data collection, relational database design, SQL implementation, Python-based analytics, and business intelligence reporting, this project demonstrates a comprehensive data analytics workflow for deriving meaningful insights from mental health survey data. The findings contribute to a better understanding of the factors associated with mental wellbeing and highlight the potential of data-driven approaches in supporting mental health awareness and decision-making.

2. Problem Statement and Objectives

Problem Statement

Mental health issues such as stress, anxiety, burnout, poor sleep quality, and emotional exhaustion have become increasingly prevalent among students and working professionals. Although surveys provide valuable information regarding mental wellbeing, raw survey responses alone do not generate actionable insights. There is a need for a structured approach that transforms survey data into meaningful information through effective database management, data analysis, and visualization techniques.

This project addresses this challenge by developing an end-to-end analytics framework that collects mental health data through a Google Form survey, organizes the responses using a normalized relational database in PostgreSQL, performs exploratory data analysis using Python, and presents key findings through interactive Power BI dashboards. The aim is to identify patterns and relationships among demographic characteristics, lifestyle habits, and mental health indicators to support data-driven understanding of mental wellbeing.

Objectives

The objectives of this project are:

- Design and distribute a Google Form survey to collect data related to demographics, lifestyle habits, and mental health indicators.
- Export, clean, and preprocess the collected survey responses to ensure data consistency, accuracy, and completeness.
- Develop a normalized relational database schema by organizing the dataset into Respondent, Lifestyle_Habits, and Mental_Health_Assessment entities.
- Establish One-to-Many (1:N) relationships using primary and foreign key constraints to maintain referential integrity within PostgreSQL.
- Implement the database in PostgreSQL by creating tables, importing cleaned datasets, and validating successful data integration through SQL queries.
- Perform exploratory data analysis (EDA) using Python to investigate trends, distributions, and relationships among variables such as stress levels, burnout, anxiety, sleep quality, work pressure, physical activity, screen time, and social support.
- Apply data preprocessing and feature engineering techniques to prepare the dataset for meaningful analysis.
- Create informative visualizations to identify key mental health patterns and factors associated with wellbeing outcomes.
- Develop interactive Power BI dashboards that provide KPI monitoring, demographic analysis, lifestyle insights, and mental health trend exploration through filters and slicers.
- Generate actionable insights that contribute to understanding the factors influencing mental wellbeing and demonstrate the practical application of data analytics techniques in healthcare-related domains.

3. Dataset Description

The Excel workbook contains three sheets with identical row counts. Each respondent has one demographic record, one lifestyle habit record, and one mental health assessment record. The notebook merges these three sheets on Respondent_ID, resulting in a 1565 row by 33 column dataframe.

Sheet	Rows	Columns	Attributes
Respondent	1565	5	Respondent_ID, Age, Gender, State, Role
Lifestyle_Habits	1565	11	Lifestyle_ID, Respondent_ID, Work_Study_Hrs, Sleep_Hrs, Physical_Activity_Days, Screen_Time, Outings_Freq, Sleep_Quality, SocialMedia_Mood, Regular_Breaks, Has_Coping_Strategy
Mental_Health_Assessment	1565	19	Assessment_ID, Respondent_ID, Low_Interest, Depressed, Sleep_Trouble, Low_Energy, Poor_Concentration, Anxiety, Emotional_Drain, Stress_Level, Burnout, Work_Pressure, WL_Balance, Life_Control, Family_Support, Coping_Mechanism, Therapist_Consult, Social_Support, MH_Affected

The respondent age ranges from 16 to 60 years, with a mean age of 27.79 and a median of 25.0. The gender distribution contains 815 male respondents and 750 female respondents. Role distribution is led by students (769), followed by employed respondents (601) and unemployed respondents (195).

Top State	Respondent Count
Andhra Pradesh	206
Karnataka	183
Telangana	160
Tamil Nadu	145
Uttar Pradesh	121
Chhattisgarh	104
Maharashtra	90
Kerala	81
Delhi	62
Rajasthan	59

4. Database Design and ER Diagram

The database was designed using normalization principles to organize the survey data into logically distinct entities and minimize data redundancy. The original survey dataset collected through Google Forms was divided into three related tables based on the nature of the information captured.

The Respondent entity stores demographic information that remains relatively stable for each participant, including attributes such as age, gender, state, and role. The Lifestyle_Habits entity contains behavioral and lifestyle-related factors, including work or study hours, sleep patterns, physical activity levels, screen time, and coping habits, which can be analyzed to understand their influence on mental wellbeing. The Mental_Health_Assessment entity records psychological indicators and support-related attributes, such as stress levels, burnout, anxiety, emotional wellbeing, coping mechanisms, and mental health support factors.

A One-to-Many (1:N) relationship was established between the Respondent table and both the Lifestyle_Habits and Mental_Health_Assessment tables using Respondent_ID as the linking attribute. This relational structure ensures data integrity through primary and foreign key constraints while supporting efficient querying and analysis of relationships between demographic characteristics, lifestyle behaviors, and mental health outcomes.

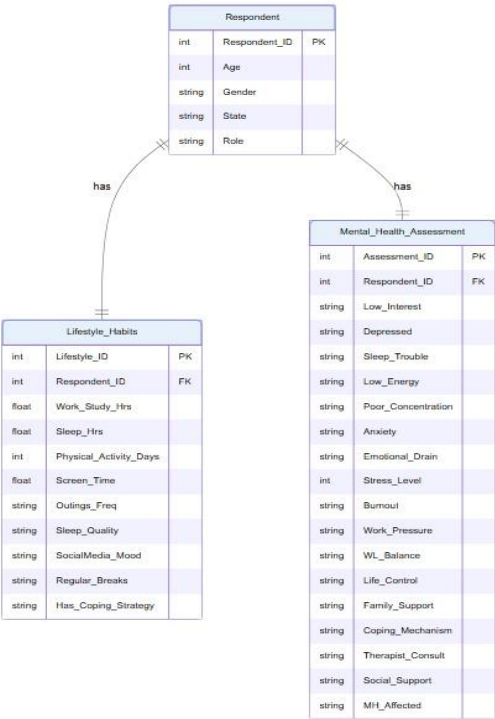


Figure 1. ER diagram showing Respondent, Lifestyle_Habits, and Mental_Health_Assessment entities.

Entity	Primary Key	Main Attributes	Purpose
Respondent	Respondent_ID	Age, Gender, State, Role	Parent demographic table
Lifestyle_Habits	Lifestyle_ID	Respondent_ID, Work_Study_Hrs, Sleep_Hrs, Physical_Activity_Days, Screen_Time, Outings_Freq, Sleep_Quality, SocialMedia_Mood, Regular_Breaks, Has_Coping_Strategy	Lifestyle child table
Mental_Health_Assessment	Assessment_ID	Respondent_ID, Low_Interest, Depressed, Sleep_Trouble, Low_Energy, Poor_Concentration, Anxiety, Emotional_Drain, Stress_Level, Burnout, Work_Pressure, WL_Balance, Life_Control, Family_Support, Coping_Mechanism, Therapist_Consult, Social_Support, MH_Affected	Assessment child table

The ER diagram illustrates Respondent as the parent entity, with Lifestyle_Habits and Mental_Health_Assessment functioning as child entities. The Respondent_ID attribute serves as the primary key in the Respondent table and is referenced as a foreign key in both child tables to establish One-to-Many (1:N) relationships.

This relational structure enforces referential integrity, ensuring that lifestyle habit records and mental health assessment records can only exist for respondents who are present in the Respondent table. As a result, the database maintains consistency, prevents orphan records, and supports reliable querying and analysis across demographic, lifestyle, and mental health dimensions.

5. PostgreSQL Implementation

The PostgreSQL implementation involved creating a database in pgAdmin and designing three normalized tables: Respondent, Lifestyle_Habits, and Mental_Health_Assessment. The Respondent table was created first because the other two tables depended on Respondent_ID as a foreign key. During implementation, datatype mismatches were encountered while importing the Lifestyle_Habits dataset because some fields contained categorical text values (e.g., "More than 8 hours", "1-2 days"). These columns were modified to VARCHAR to ensure successful import. After creating the tables, the cleaned Excel sheets were exported as CSV files and imported into their respective PostgreSQL tables using pgAdmin's Import/Export Data option. Finally, validation queries such as SELECT COUNT(*) were executed to confirm that all 1,565 records were successfully imported into each table.


```
CREATE TABLE Respondent (  
    Respondent_ID INT PRIMARY KEY,  
    Age INT,  
    Gender VARCHAR(20),  
    State VARCHAR(100),  
    Role VARCHAR(100)  
);  
  
CREATE TABLE Lifestyle_Habits (  
    Lifestyle_ID INT PRIMARY KEY,  
    Respondent_ID INT REFERENCES Respondent(Respondent_ID),  
    Work_Study_Hrs VARCHAR(50),  
    Sleep_Hrs VARCHAR(50),  
    Physical_Activity_Days VARCHAR(50),  
    Screen_Time VARCHAR(50),  
    Outings_Freq VARCHAR(50),  
    Sleep_Quality INT,  
    SocialMedia_Mood INT,  
    Regular_Breaks VARCHAR(10),  
    Has_Coping_Strategy VARCHAR(10)  
);  
  
CREATE TABLE Mental_Health_Assessment (  
    Assessment_ID INT PRIMARY KEY,  
    Respondent_ID INT REFERENCES Respondent(Respondent_ID),  
    Low_Interest VARCHAR(20),  
    Depressed VARCHAR(20),  
    Sleep_Trouble VARCHAR(20),  
    Low_Energy VARCHAR(20),  
    Poor_Concentration VARCHAR(20),  
    Anxiety VARCHAR(20),  
    Emotional_Drain VARCHAR(20),  
    Stress_Level INT,  
    Burnout VARCHAR(20),  
    Work_Pressure VARCHAR(20),  
    WL_Balance VARCHAR(20),  
    Life_Control VARCHAR(20),  
    Family_Support VARCHAR(20),  
    Coping_Mechanism VARCHAR(100),  
    Therapist_Consult VARCHAR(20),  
    Social_Support VARCHAR(20),  
    MH_Affected VARCHAR(20)  
);
```

6. Data Cleaning and Preprocessing

The original Excel dataset downloaded from the Google Forms survey contained all respondent information in a single worksheet. To improve data organization and support relational database design, the dataset was cleaned and divided into three separate worksheets: Respondent, Lifestyle_Habits, and Mental_Health_Assessment. These worksheets were created based on the logical grouping of demographic, lifestyle, and mental health attributes while maintaining relationships through Respondent_ID.

The Python notebook loaded the cleaned Excel workbook, displayed the sheet names, read the three worksheets, and merged them using Respondent_ID to create a unified analytical dataset. The merged dataframe contained 1,565 rows and 33 columns. Data inspection procedures included examining the head and tail of the dataset, generating random samples, checking shape and data types, reviewing summary statistics, identifying unique values, assessing missing values, and detecting duplicate records.

Quality Check	Result
Merged rows	1565
Merged analytical columns	33
Missing values	0
Duplicate records	0
Numeric attributes after merge	19
String/Categorical attributes after merge	14

Since the dataset contained no missing values or duplicate records, the preprocessing stage focused primarily on schema validation, categorical value ordering, score interpretation, and feature engineering. Additional derived features such as Age_Group, Mental_Health_Score, numeric encoding for MH_Affected, and ordered categories for Sleep_Hrs, Work_Study_Hrs, Physical_Activity_Days, and Screen_Time were created to support exploratory data analysis and dashboard development.

Metric	Average Score
Stress_Level	2.828
Burnout	2.785
Work_Pressure	2.842
WL_Balance	3.309
Life_Control	3.417
Family_Support	3.752
Sleep_Quality	2.981
SocialMedia_Mood	2.935
Anxiety	2.610
Depressed	2.662

7. Exploratory Data Analysis (EDA) and Visualization Using Python

After data cleaning and validation, Exploratory Data Analysis (EDA) was performed using Pandas for data manipulation and Matplotlib and Seaborn for visualization. The analysis explored respondent demographics, lifestyle habits, and mental health indicators through various visualizations, including count plots for respondent composition, histograms for age distribution, heatmaps for categorical relationships, line plots for age-based trends, box and violin plots for distribution comparisons, scatter plots for examining variable relationships, and correlation heatmaps for identifying associations among numeric variables.

The EDA revealed several important findings. The average Stress Level was 2.83, Burnout was 2.79, Anxiety was 2.61, Depression score was 2.66, and Sleep Quality was 2.98. Mental health was reported as affected by 479 respondents (30.61%), while only 182 respondents indicated that they had consulted a therapist. Further analysis showed that employed respondents exhibited the highest average Stress Level (3.05) and Burnout (2.96) scores. These findings highlight the significant relationship between work pressure and burnout and provided the foundation for developing the interactive Power BI dashboards.

8. Detailed Interpretation of Notebook Visuals

This chapter includes every extracted graph from the Jupyter Notebook. Each figure is the actual PNG output saved inside the notebook, linked to the code cell that produced it. The interpretation combines the visible chart purpose with computed dataset statistics.

8.1 Average Stress Level Trend Across Ages

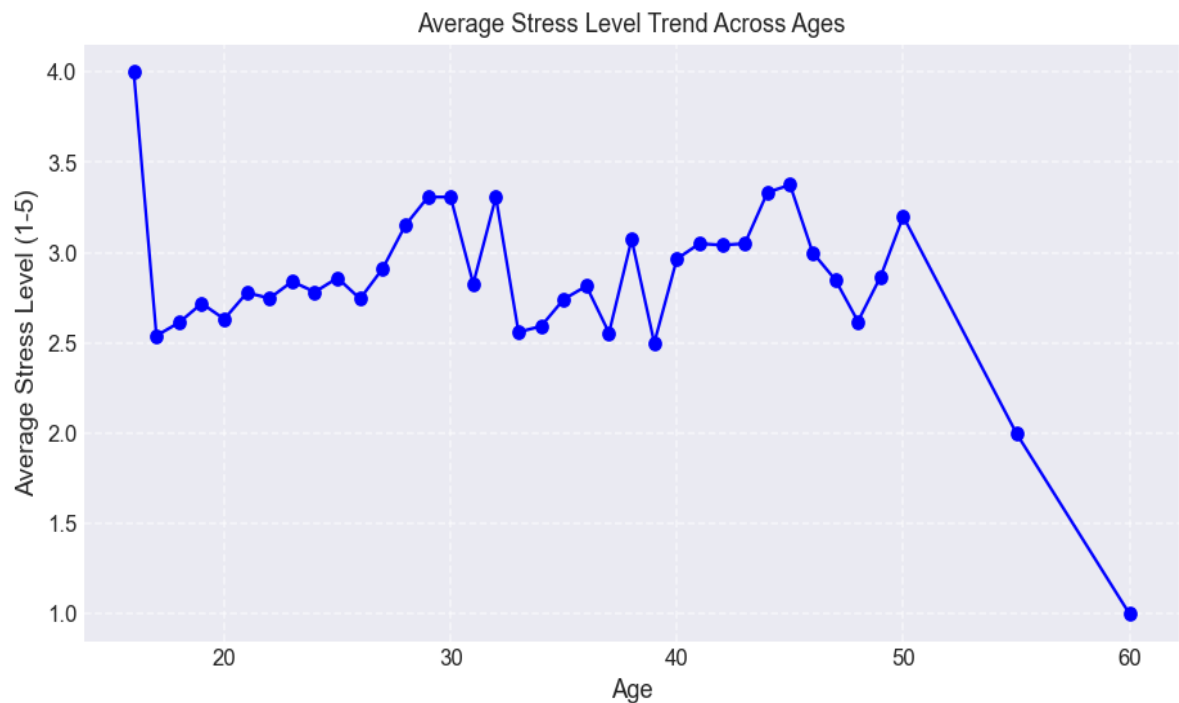


Figure 4. Average Stress Level Trend Across Ages.

Methodology: The notebook groups records by Age and calculates mean Stress_Level. A Matplotlib line plot with markers visualizes how average stress changes across ages.

What the graph represents: The chart is useful for identifying age bands where stress is elevated. Because the dataset includes respondents from 16 to 60 years, the line makes age-related variation visible while smoothing individual-level noise.

Significance: In the mental health context, age-linked stress variation helps separate student pressures, early-career pressures, and later-life responsibilities.

8.2 Number of Respondents by Role

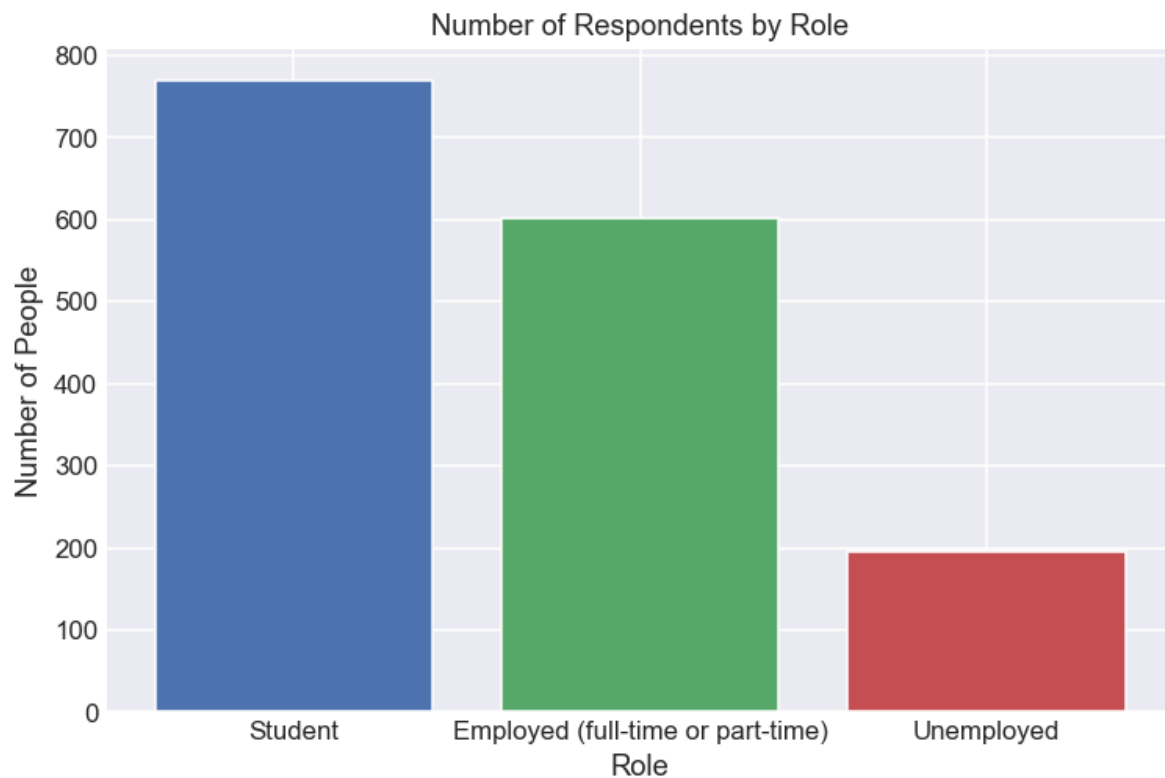


Figure 5. Number of Respondents by Role.

Methodology: The notebook uses `value_counts` on `Role` and plots the counts as a bar chart.

What the graph represents: Students are the largest respondent group, followed by employed respondents and then unemployed respondents. This composition affects interpretation because aggregate trends are influenced by student and employed experiences.

Significance: Role segmentation is essential because workload, study burden, and employment pressure are not identical stressors.

8.3 Sleep Quality vs Stress Level

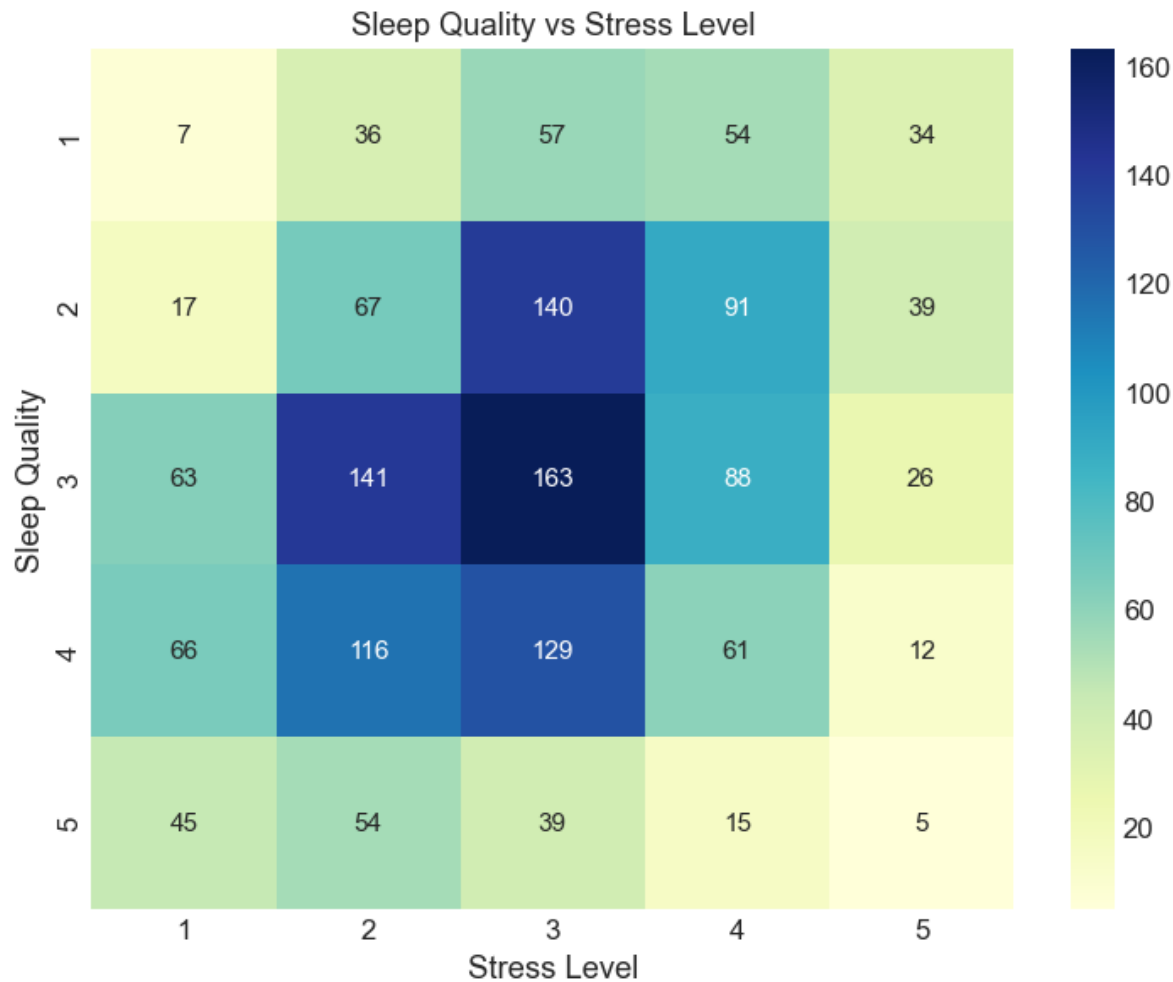


Figure 6. Sleep Quality vs Stress Level.

Methodology: A crosstab of Sleep_Quality and Stress_Level is visualized as a Seaborn heatmap.

What the graph represents: The computed relationship shows average stress decreasing from 3.38 at sleep quality 1 to 2.25 at sleep quality 5.

Significance: This supports the wellbeing insight that better sleep quality is associated with lower stress.

8.4 Age Distribution of Respondents

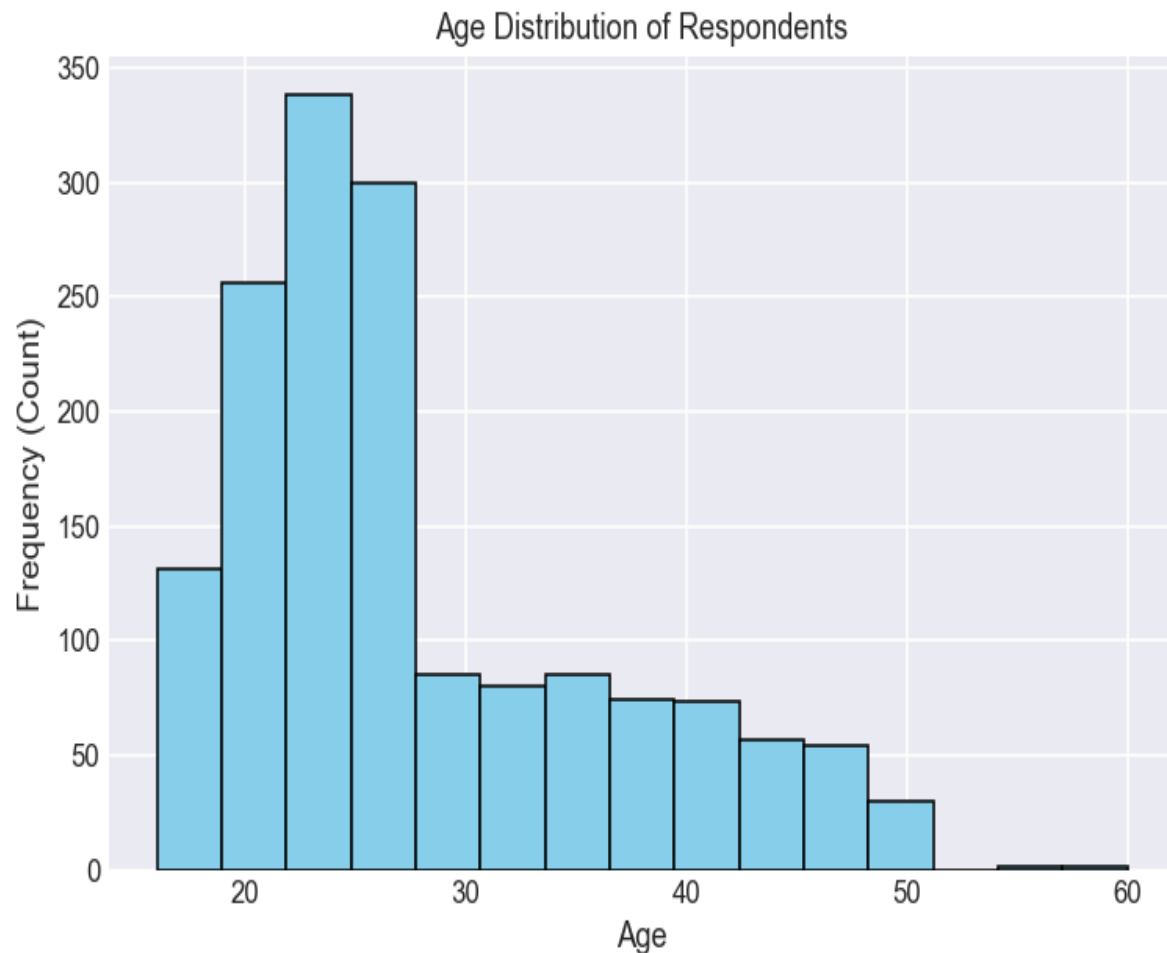


Figure 7. Age Distribution of Respondents.

Methodology: The notebook plots a histogram of Age using 15 bins.

What the graph represents: The age profile is concentrated in younger adult groups, with mean age 27.79 and median age 25.0.

Significance: A young sample explains why student lifestyle, screen time, and study pressure are important analytical dimensions.

8.5 Proportion of Respondents with Affected Mental Health

Proportion of Respondents with Affected Mental Health

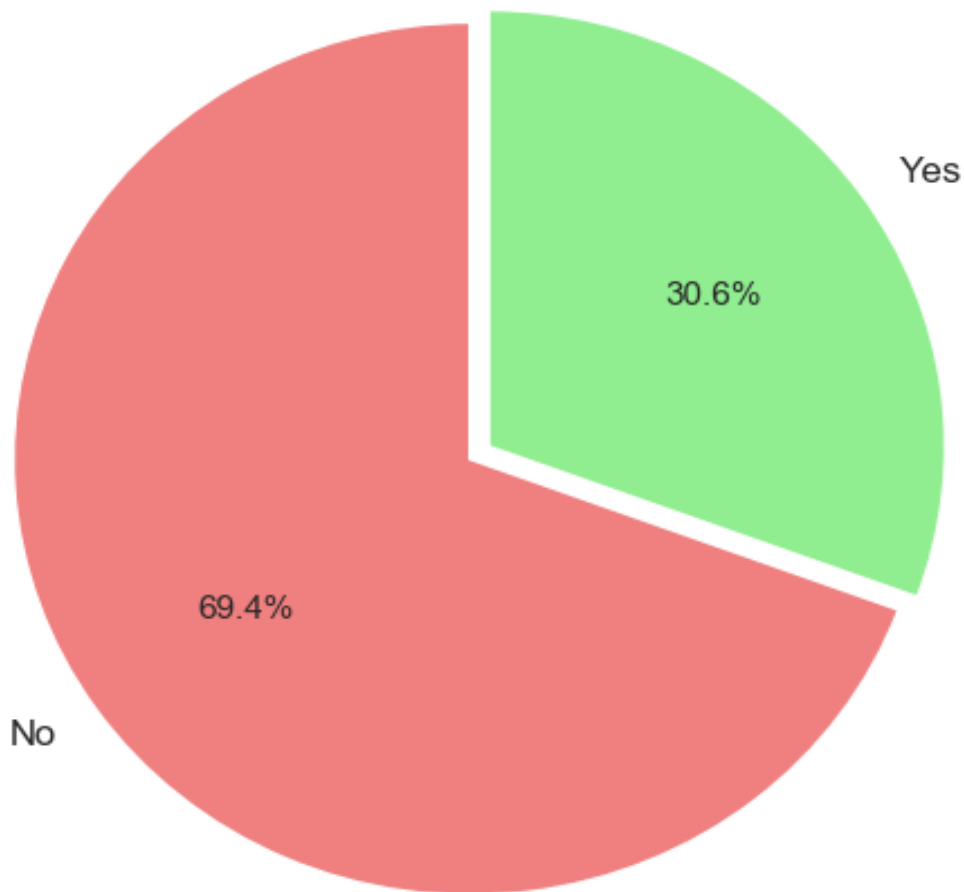


Figure 8. Proportion of Respondents with Affected Mental Health.

Methodology: The notebook uses value_counts on MH_Affected and displays a pie chart.

What the graph represents: Mental health is marked affected for 479 respondents (30.61%) and not affected for 1086 respondents.

Significance: This KPI provides the main outcome variable for mental health impact analysis.

8.6 Burnout Levels by Gender

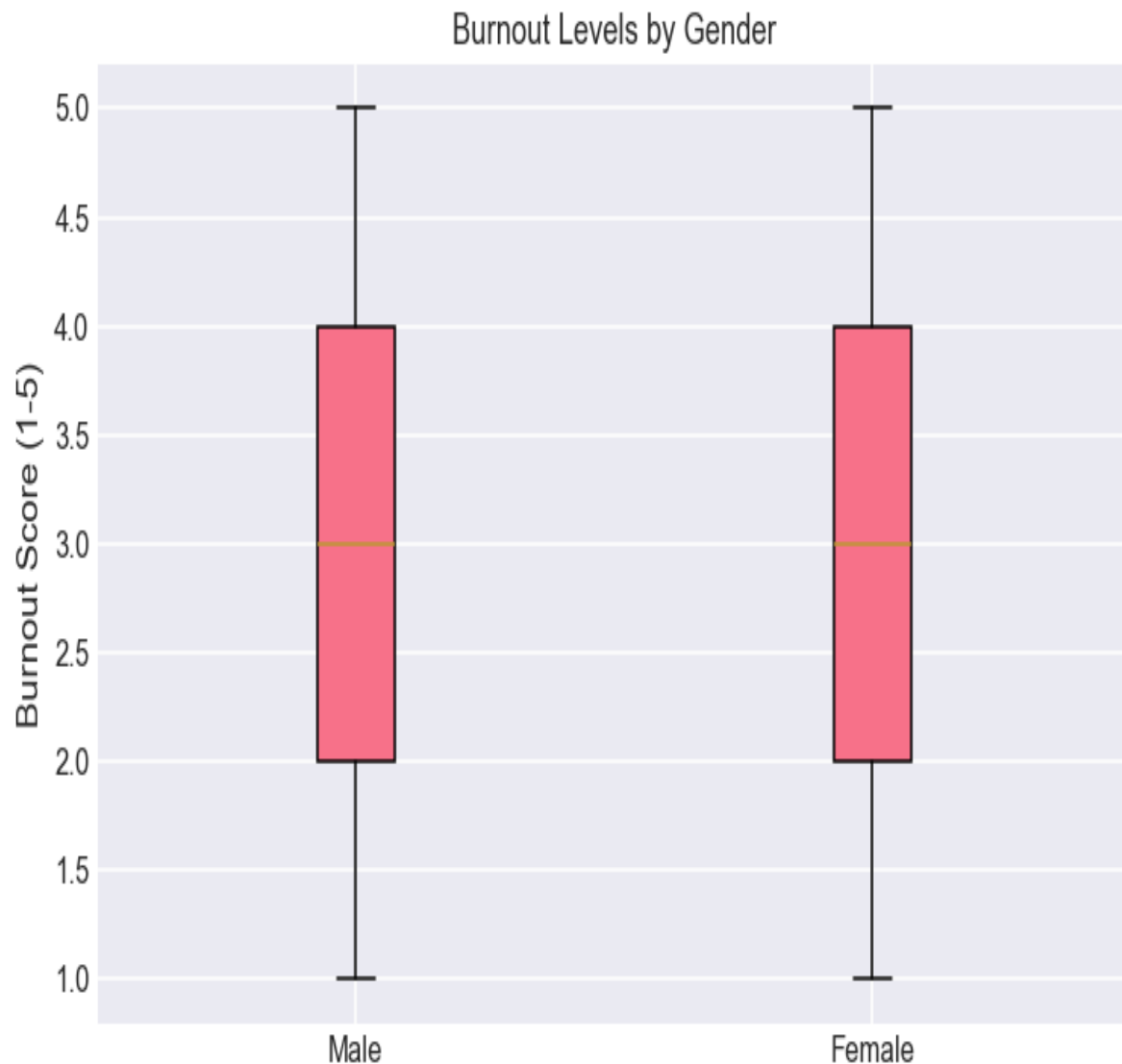


Figure 9. Burnout Levels by Gender.

Methodology: The notebook separates Burnout scores by Gender and plots a box plot.

What the graph represents: Average burnout is close across genders: female 2.82 and male 2.76.

Significance: Similar central tendencies suggest that gender alone is not the strongest differentiator in this dataset.

8.7 Distribution of Stress Levels by Gender

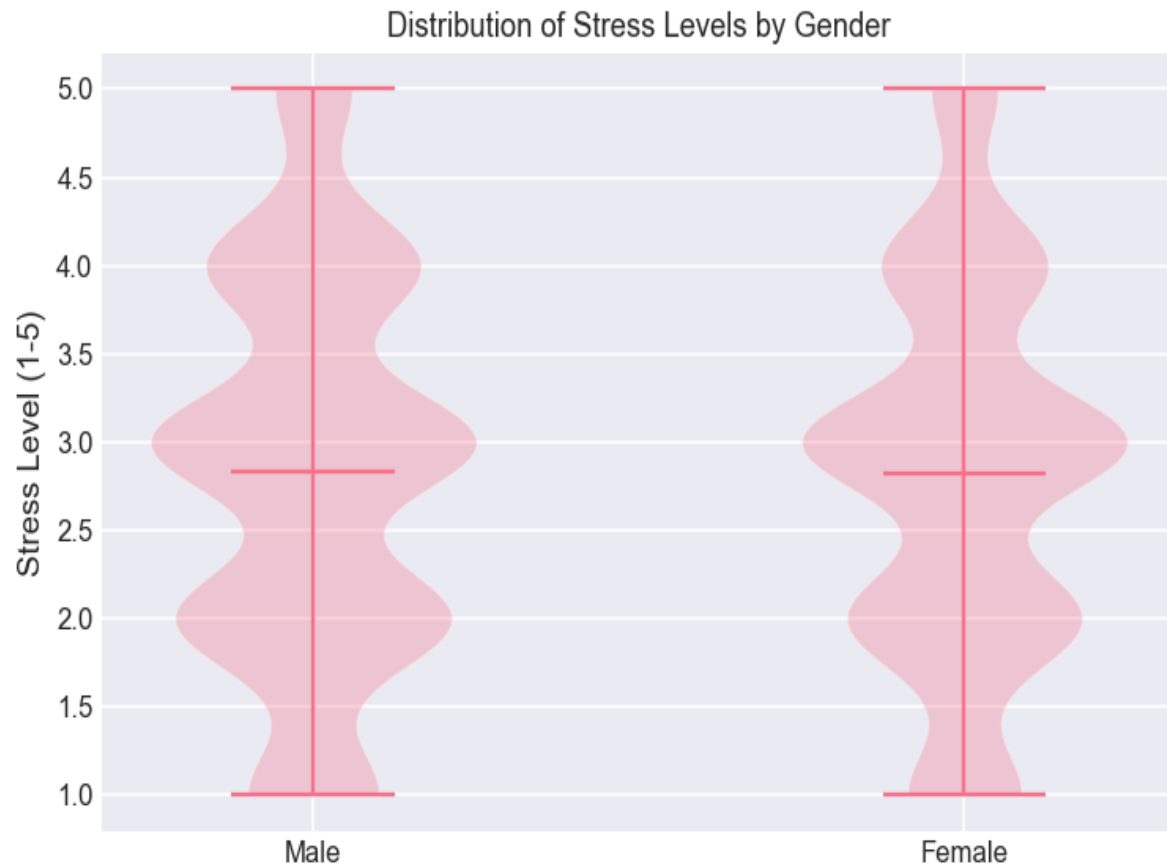


Figure 10. Distribution of Stress Levels by Gender.

Methodology: Stress_Level is split by Gender and shown through a violin plot with means.

What the graph represents: Average stress is nearly equal by gender: female 2.82 and male 2.83.

Significance: The visual emphasizes distribution shape rather than only averages, which is useful for seeing whether one group has more extreme scores.

8.8 Work Pressure vs Burnout



Figure 11. Work Pressure vs Burnout.

Methodology: A crosstab of Work_Pressure and Burnout is displayed as a red heatmap.

What the graph represents: The numeric correlation between work pressure and burnout is 0.30, showing a strong positive association.

Significance: This is one of the most important project findings: pressure and burnout rise together.

8.9 Average Work-Life Balance by Daily Work/Study Hours

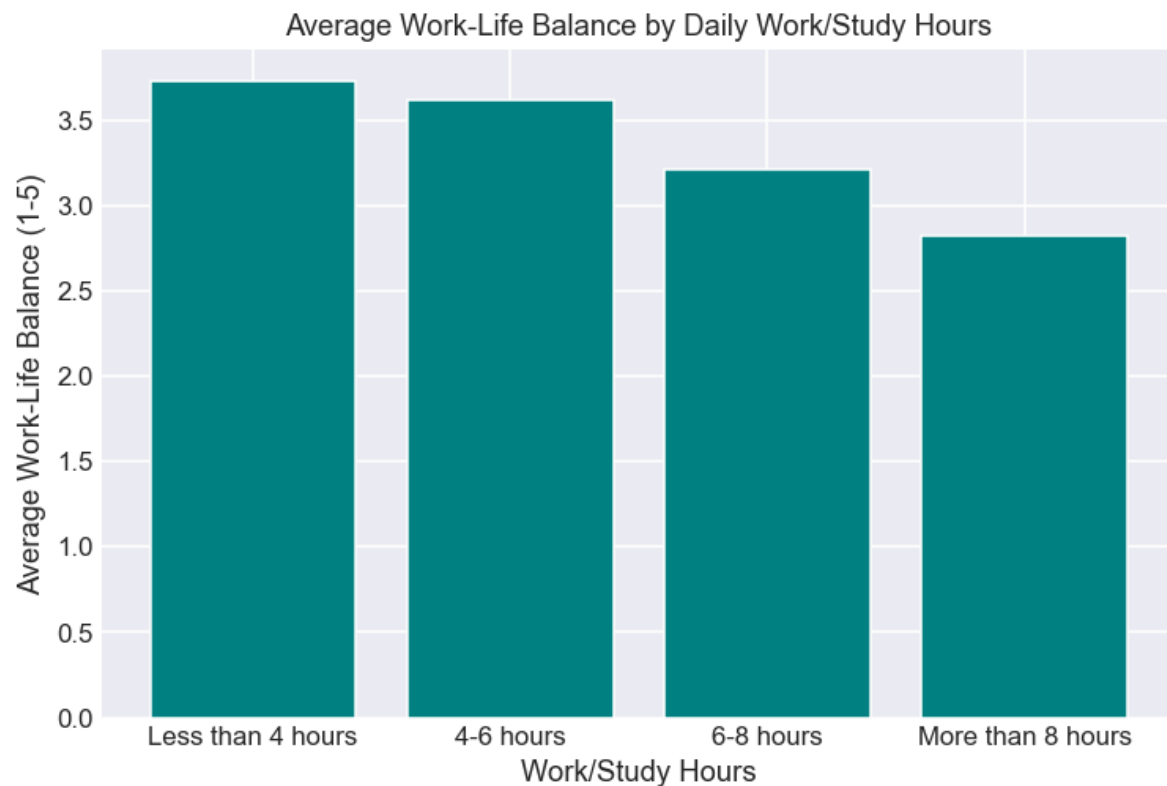


Figure 12. Average Work-Life Balance by Daily Work/Study Hours.

Methodology: Work_Study_Hrs is ordered categorically and grouped by average WL_Balance.

What the graph represents: The chart tests whether longer work or study periods are associated with lower perceived balance.

Significance: This supports intervention planning around workload management, breaks, and realistic study or work schedules.

8.10 Average Mental Health Score by Age Group

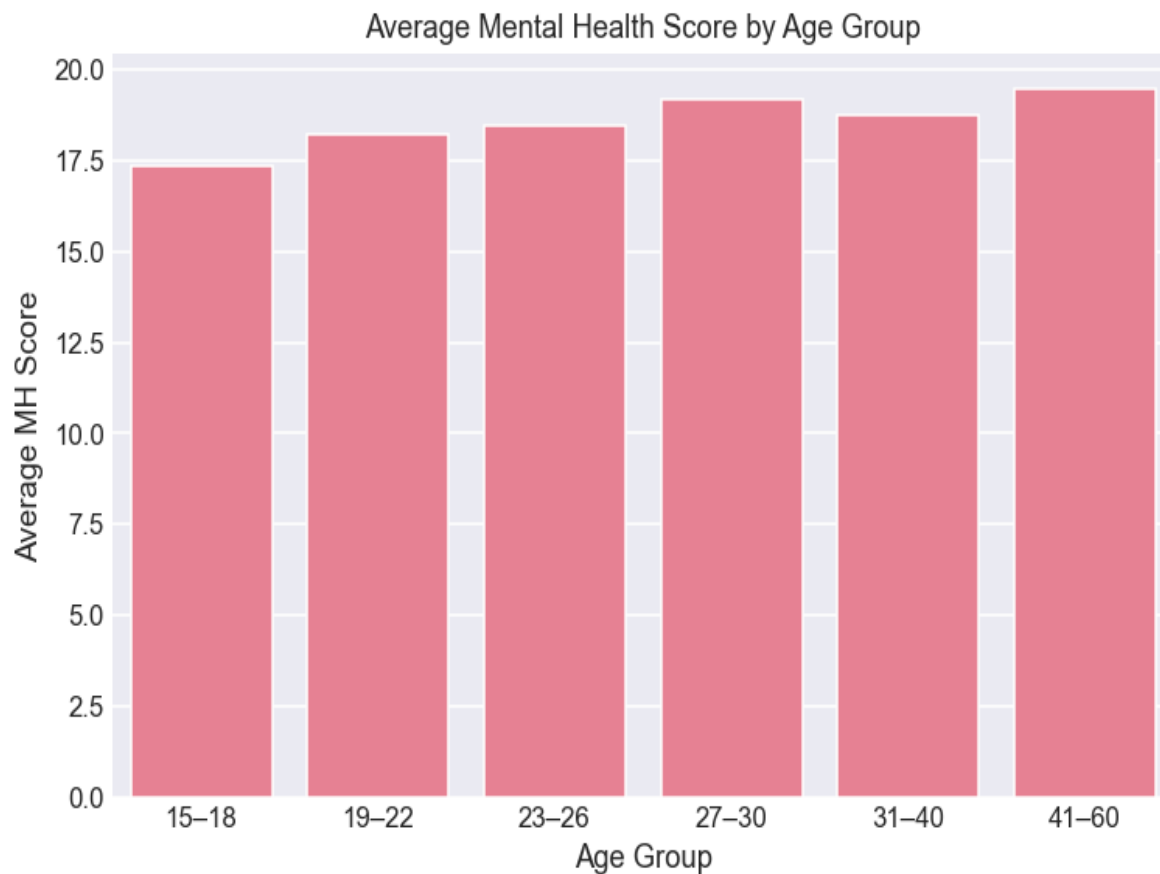


Figure 13. Average Mental Health Score by Age Group.

Methodology: The notebook creates Age_Group and averages a derived MH_Score across groups.

What the graph represents: The derived score summarizes multiple symptoms, allowing age groups to be compared on a single composite measure.

Significance: Composite scoring is useful for dashboards because it simplifies several symptom indicators into a readable mental-health burden metric.

8.11 Top 10 States by Number of Respondents

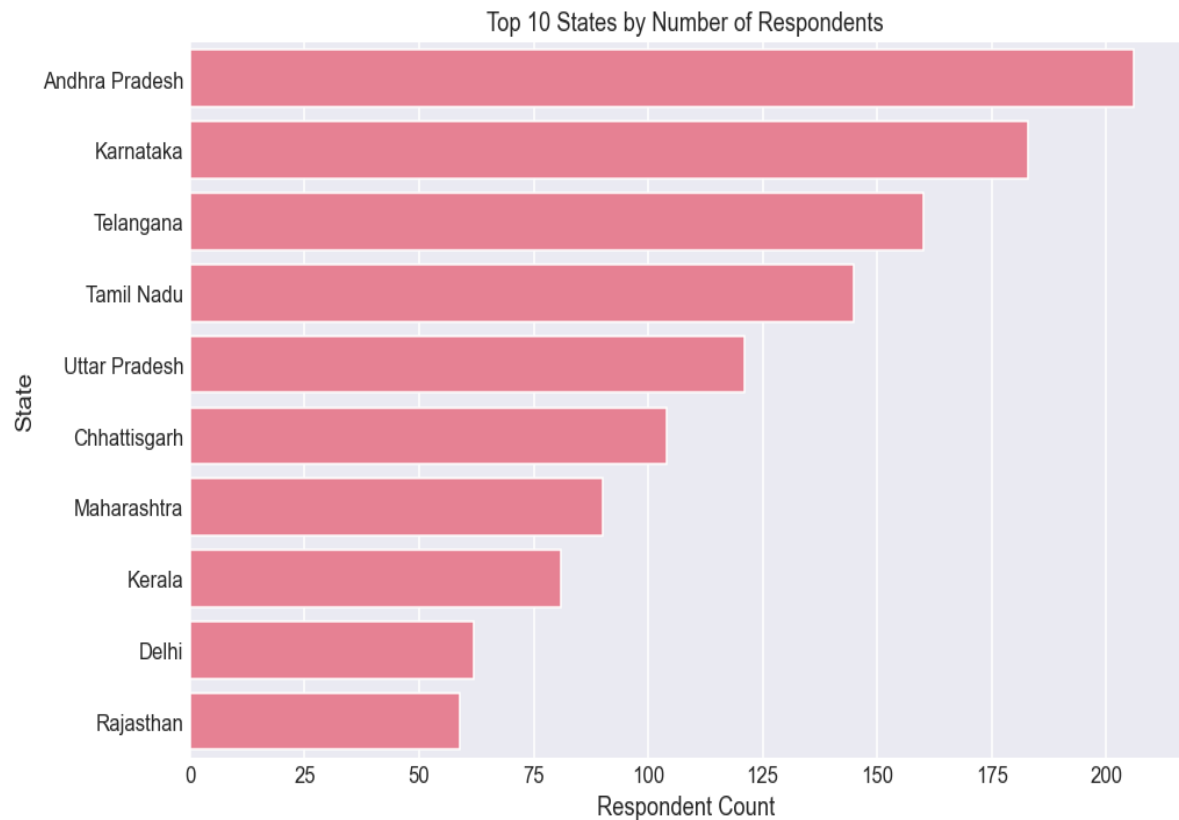


Figure 14. Top 10 States by Number of Respondents.

Methodology: The notebook counts State values and plots the ten highest respondent counts.

What the graph represents: Andhra Pradesh (206), Karnataka (183), and Telangana (160) are the largest state groups.

Significance: State distribution is important for interpreting geographic slicers in Power BI.

8.12 Top 10 States by Mental Health Impact Rate

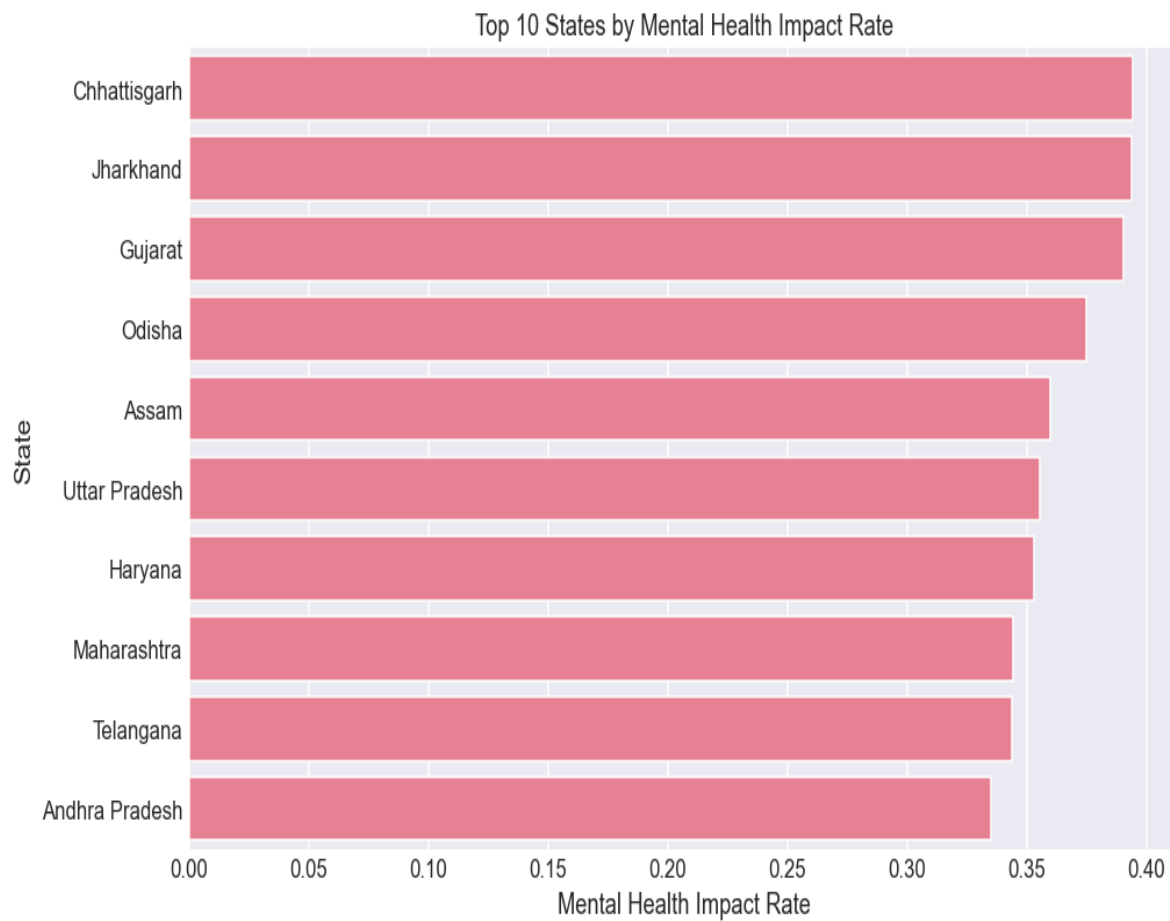


Figure 15. Top 10 States by Mental Health Impact Rate.

Methodology: The notebook encodes MH_Affected as numeric and calculates mean impact rate by State.

What the graph represents: The chart highlights states with higher affected-rate proportions rather than just higher sample size.

Significance: Rate-based state analysis is more meaningful than count-based comparison because it controls for unequal respondent totals.

8.13 Average Severity of Mental Health Symptoms

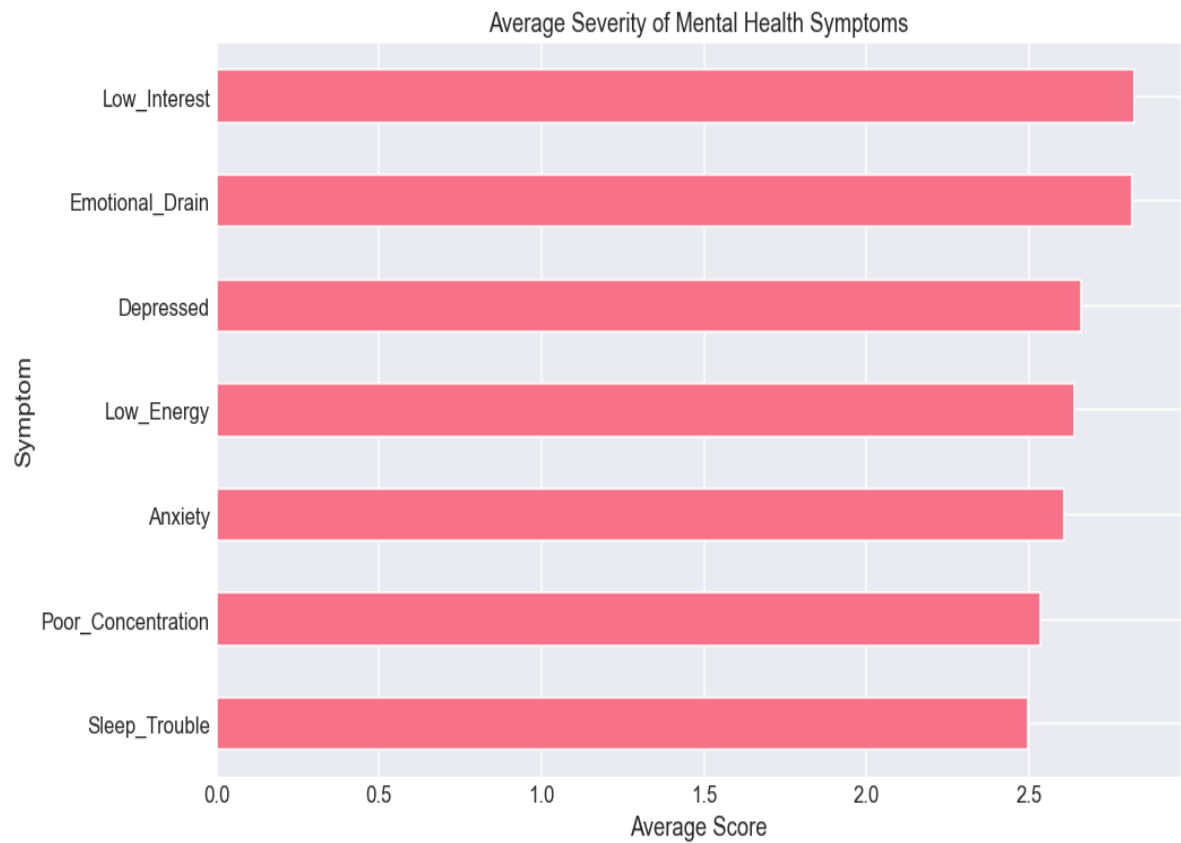


Figure 16. Average Severity of Mental Health Symptoms.

Methodology: The notebook averages seven symptom columns and plots horizontal bars.

What the graph represents: The highest average symptoms include Low_Interest (2.83) and Emotional_Drain (2.81).

Significance: Symptom ranking helps identify where awareness and support programs should focus.

8.14 Age vs Mental Health Score

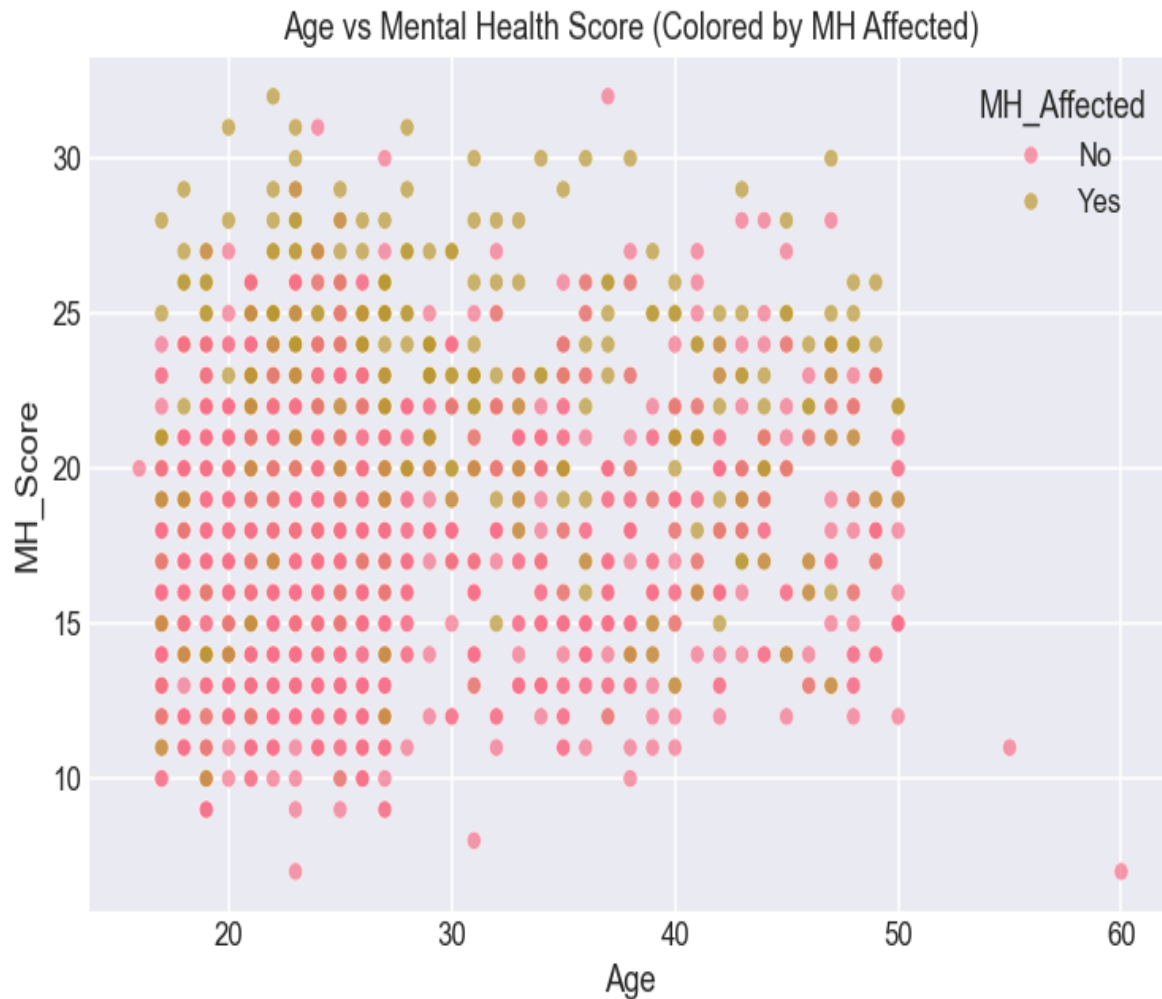


Figure 17. Age vs Mental Health Score.

Methodology: A Seaborn scatterplot maps Age to MH_Score and colors points by MH_Affected.

What the graph represents: This visual reveals individual-level dispersion and separates affected from non-affected respondents.

Significance: Scatter-level analysis prevents overreliance on group means and shows that mental health experiences vary within every age band.

8.15 Trend of Average Stress Level Across Ages

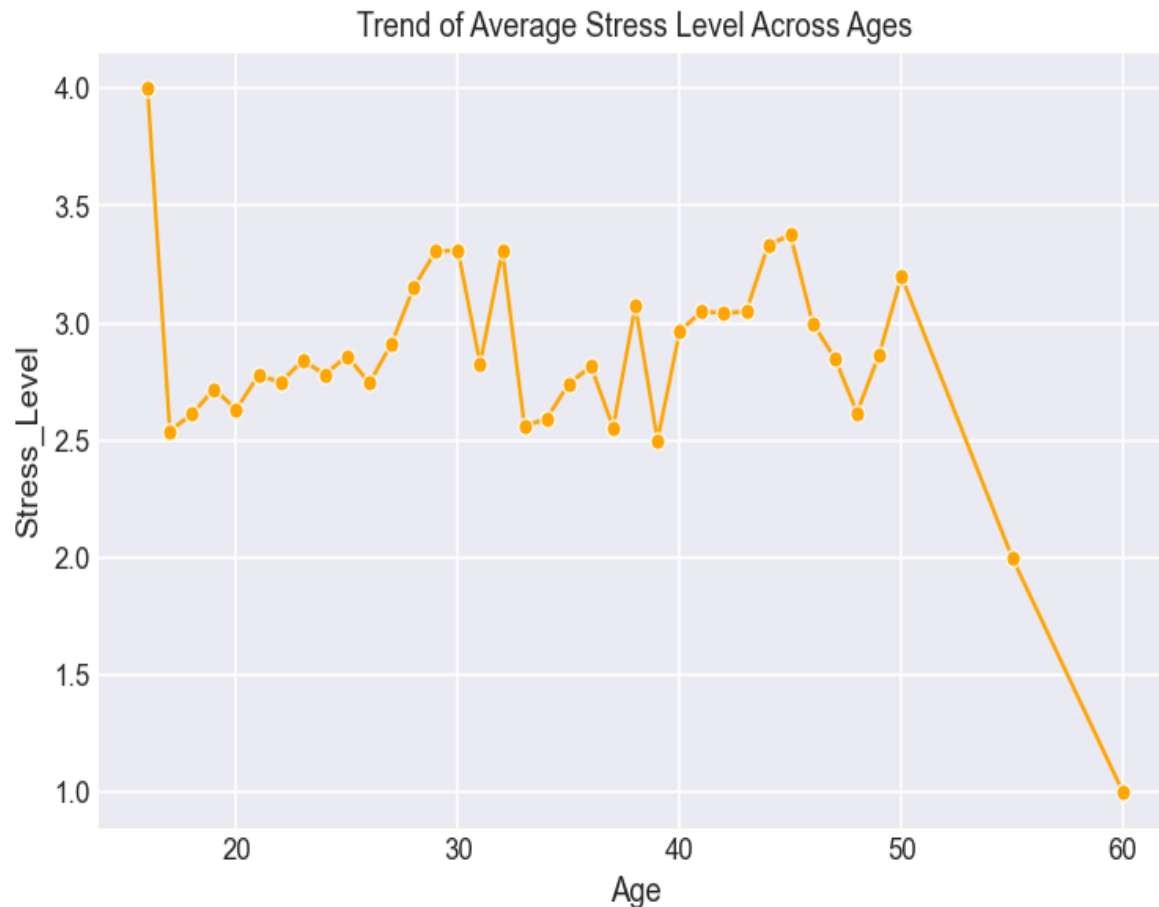


Figure 18. Trend of Average Stress Level Across Ages.

Methodology: A Seaborn lineplot estimates mean Stress_Level by Age with markers.

What the graph represents: This repeats the age-stress trend with Seaborn styling and confirms whether the pattern is stable.

Significance: Using a second plotting approach strengthens visual validation of the same relationship.

8.16 Average Burnout Score by Role

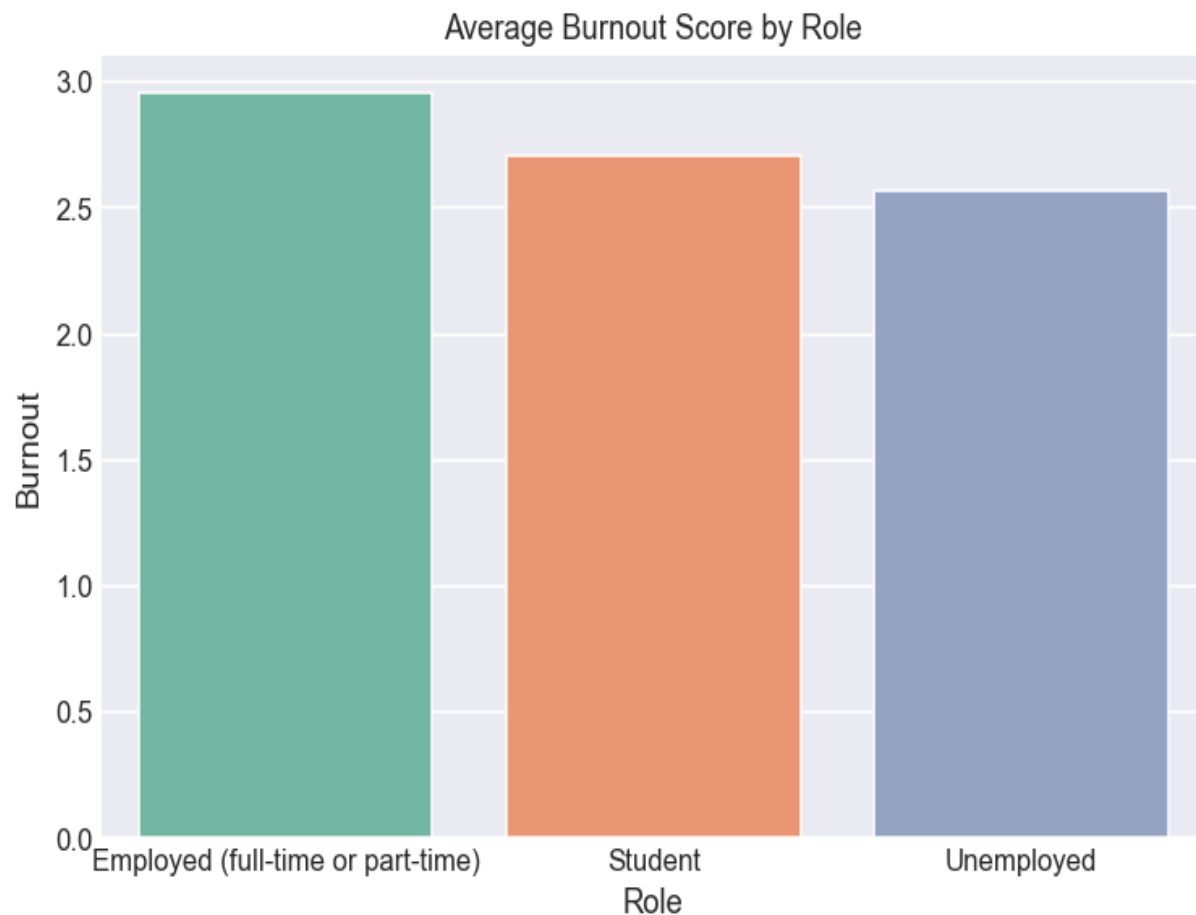


Figure 19. Average Burnout Score by Role.

Methodology: A Seaborn barplot compares Burnout by Role.

What the graph represents: Employed respondents show the highest mean burnout (2.96), followed by students (2.71).

Significance: This aligns with occupational workload and recovery-time concerns.

8.17 Count of Respondents by Sleep Hours

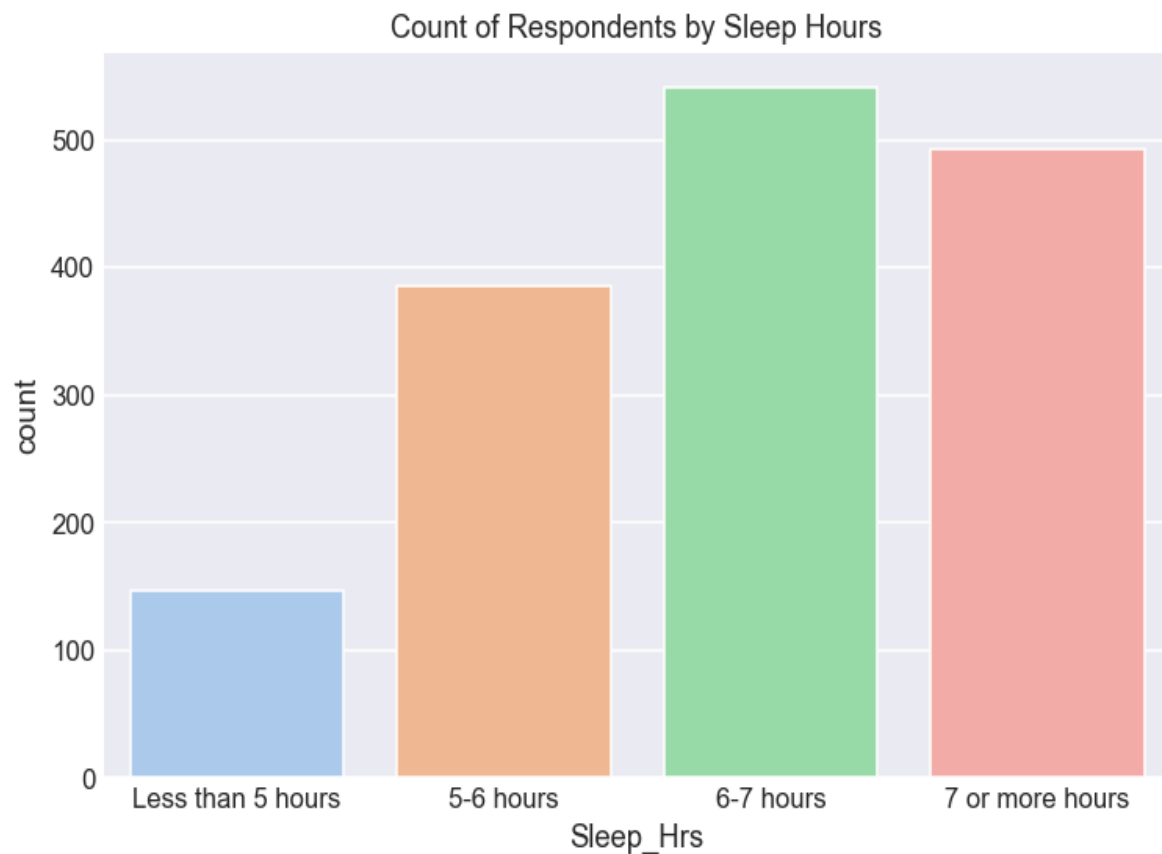


Figure 20. Count of Respondents by Sleep Hours.

Methodology: A Seaborn countplot orders Sleep_Hrs categories from low to high sleep duration.

What the graph represents: The most common sleep group is 6-7 hours (541) followed by 7 or more hours (492).

Significance: Sleep duration distribution contextualizes sleep-quality and stress findings.

8.18 Stress Levels Distributed by Physical Activity Days

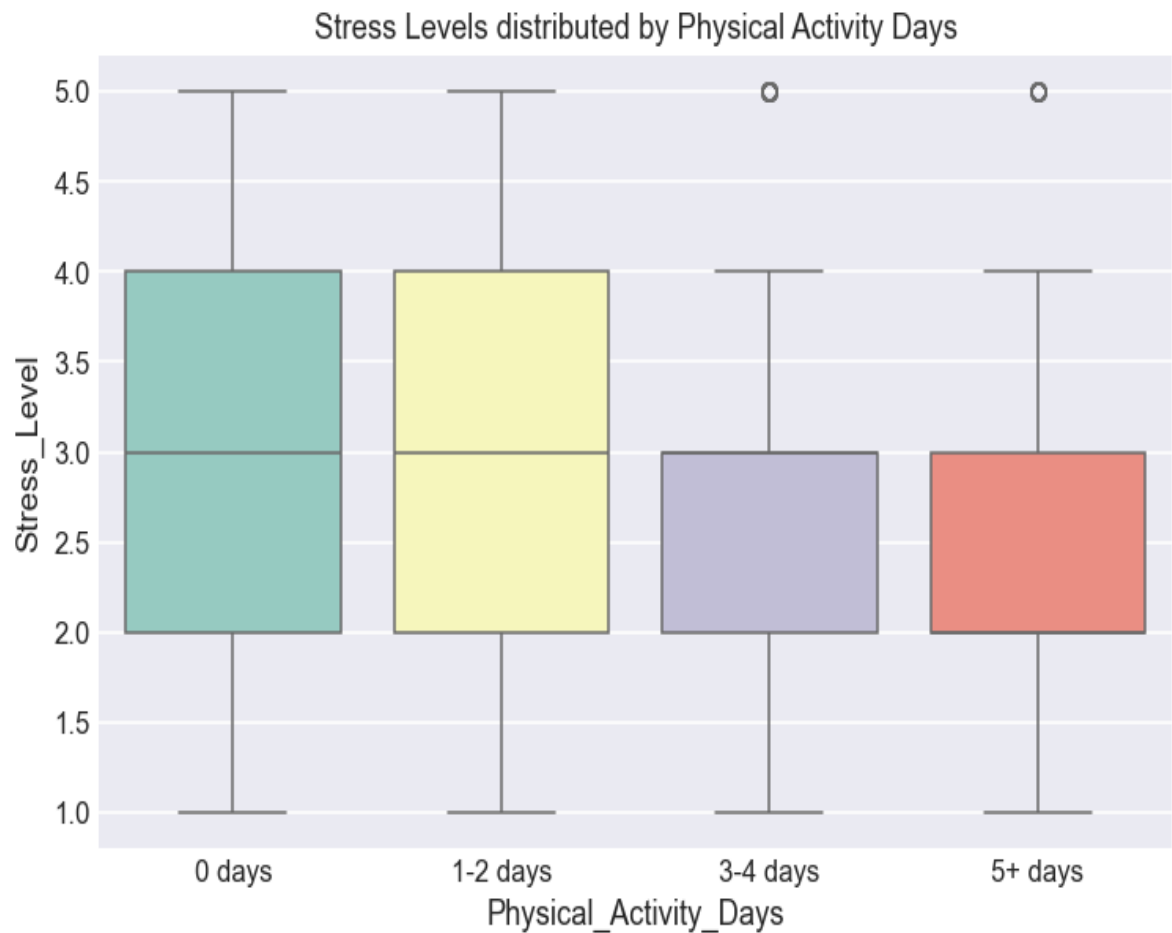


Figure 21. Stress Levels Distributed by Physical Activity Days.

Methodology: A Seaborn boxplot compares Stress_Level across physical activity frequency categories.

What the graph represents: The visual checks whether respondents with more active days show lower or more stable stress distributions.

Significance: Physical activity is a modifiable wellbeing habit, so this chart is actionable.

8.19 Mental Health Score Distribution by Social Support

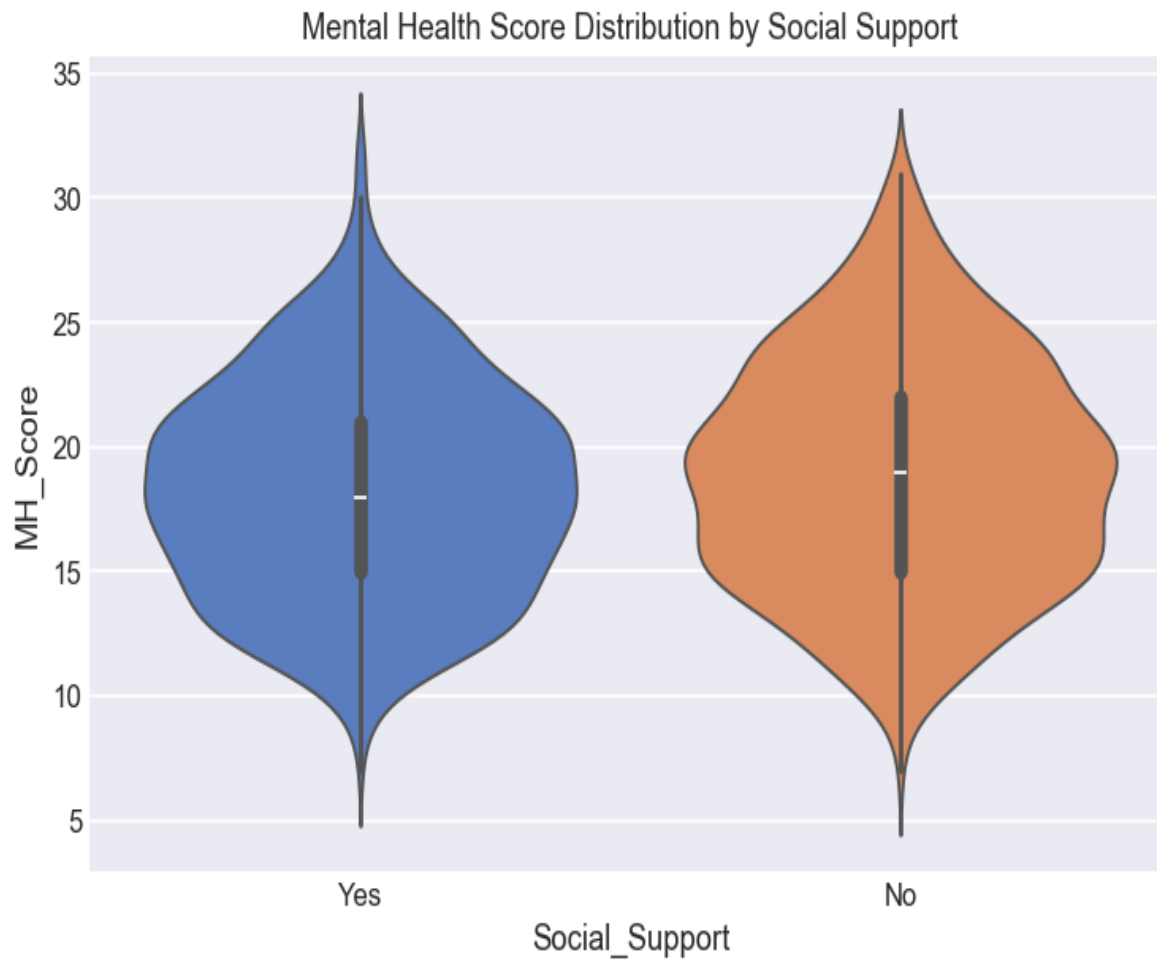


Figure 22. Mental Health Score Distribution by Social Support.

Methodology: A violin plot compares MH_Score for respondents with and without Social_Support.

What the graph represents: Social support is reported by 944 respondents and absent for 621 respondents.

Significance: The chart supports the interpretation of support systems as a protective factor.

8.20 Impact of Screen Time on Sleep Quality

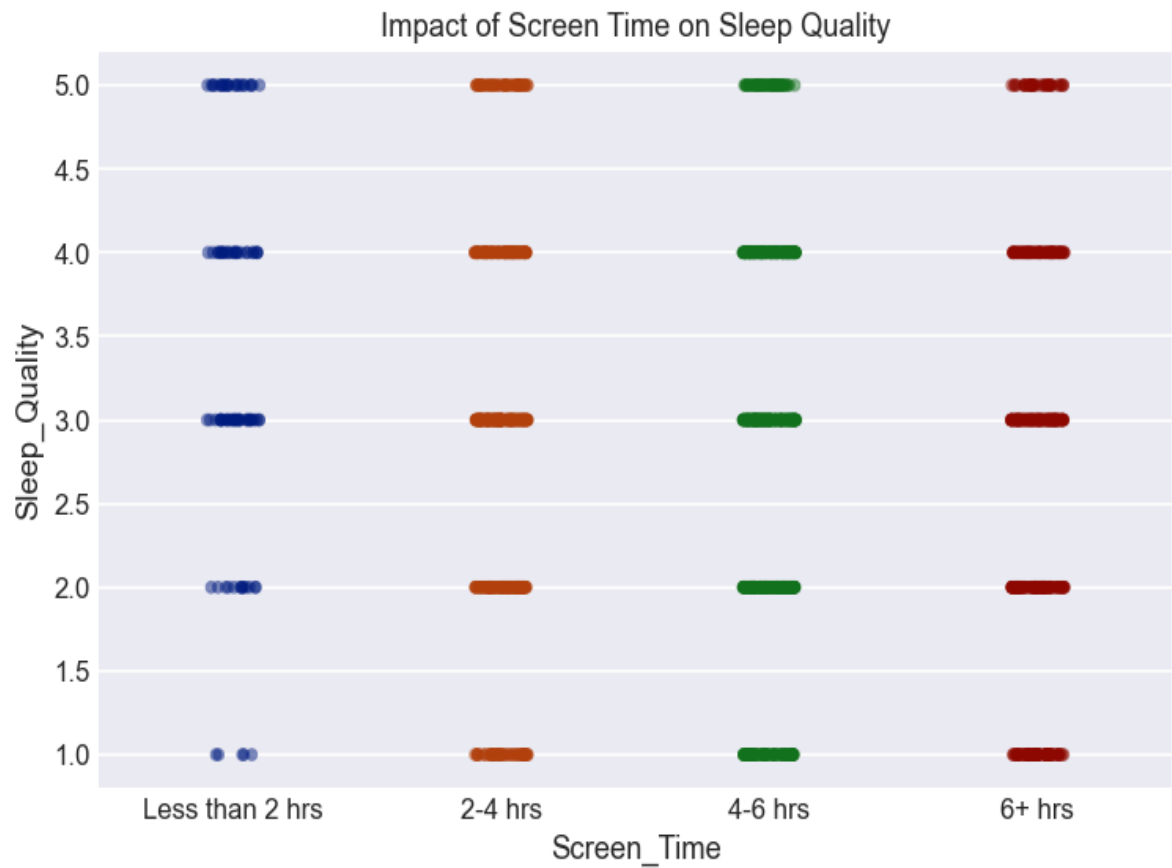


Figure 23. Impact of Screen Time on Sleep Quality.

Methodology: A strip plot maps Screen_Time categories against Sleep_Quality with jittered points.

What the graph represents: The most common screen-time category is 4-6 hrs (617) and a large group reports 6+ hrs (387).

Significance: Screen time is directly relevant to sleep hygiene, mood, and attention.

8.21 Burnout vs Regular Breaks

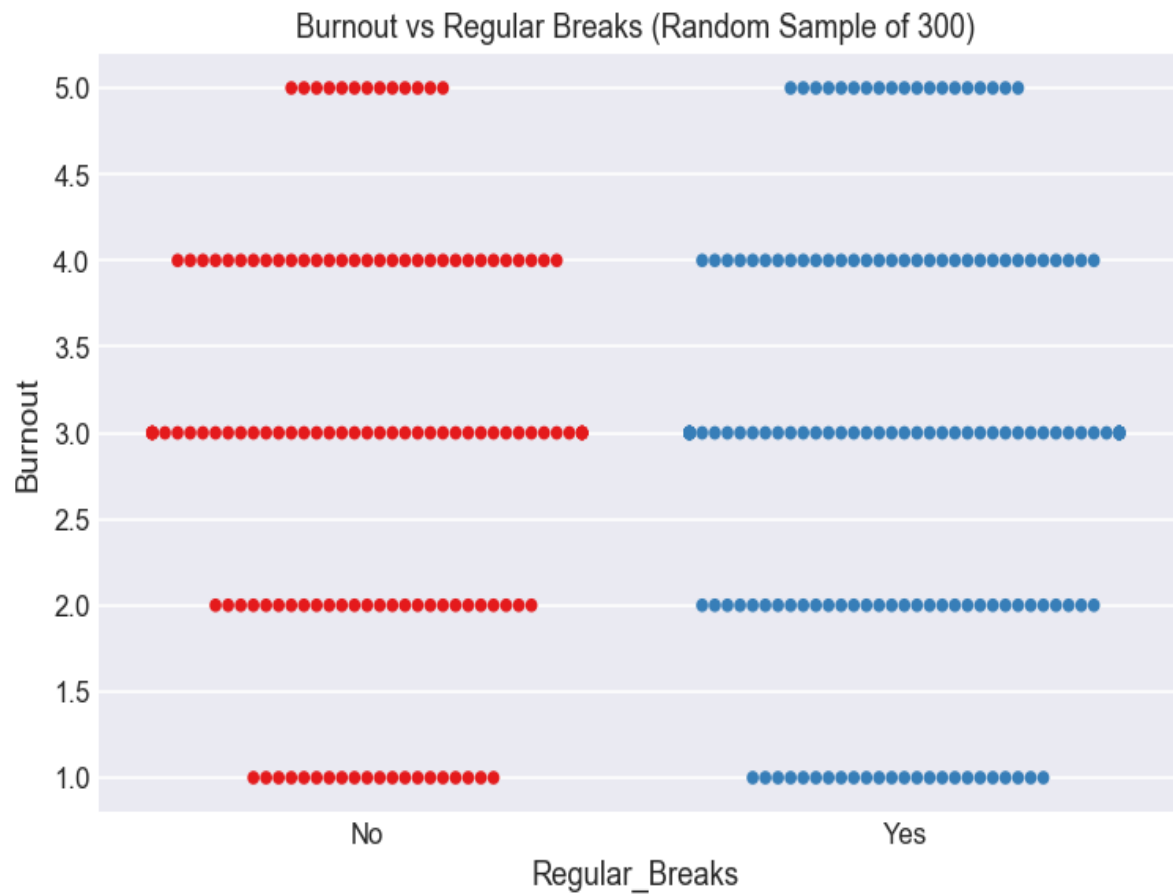


Figure 24. Burnout vs Regular Breaks.

Methodology: A random sample of 300 records is plotted with a swarmplot to reduce point overlap.

What the graph represents: Regular breaks are reported by 870 respondents and not reported by 695.

Significance: The chart turns a simple yes/no habit into a burnout comparison.

8.22 Average Life Control by Outings Frequency

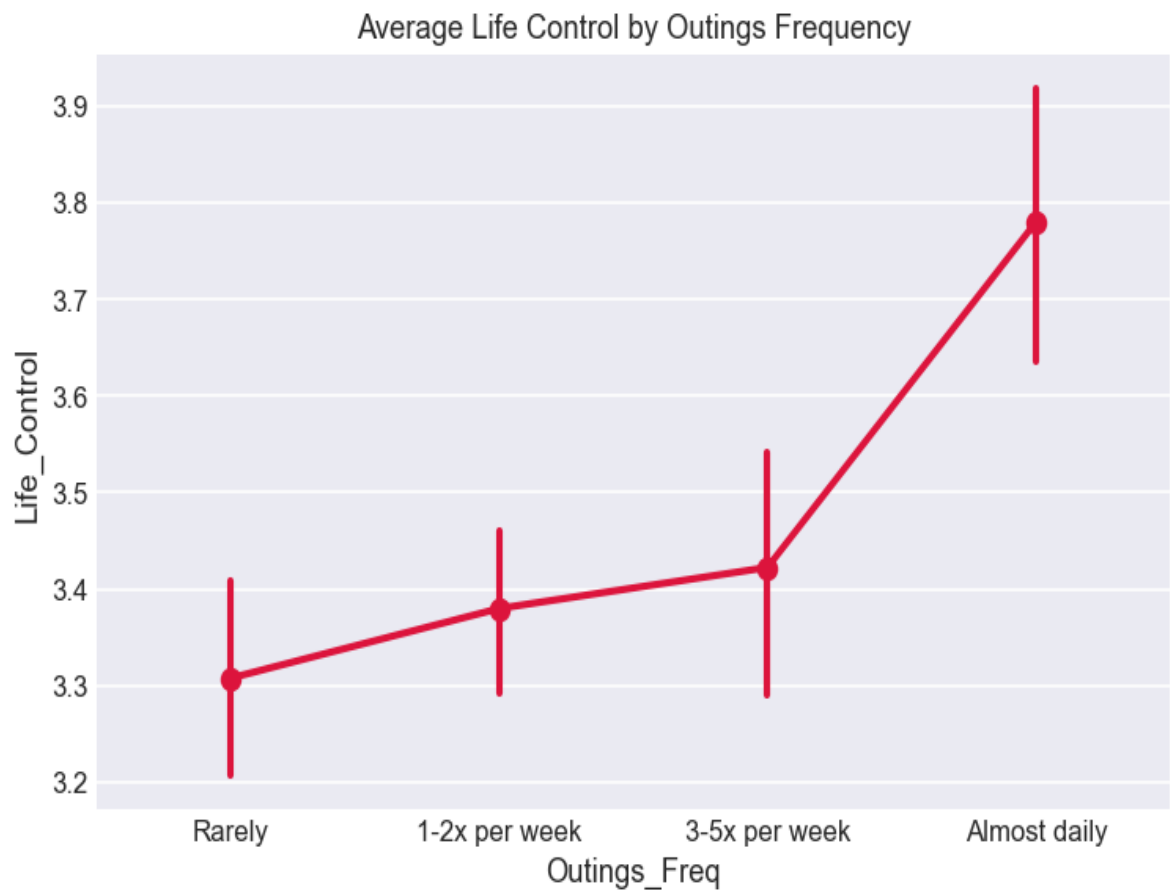


Figure 25. Average Life Control by Outings Frequency.

Methodology: A point plot orders outings frequency and computes average Life_Control.

What the graph represents: The chart tests whether social or outdoor engagement corresponds with stronger perceived control.

Significance: Life control is a resilience-related indicator and belongs with social support and coping analysis.

8.23 Matplotlib EDA Dashboard



Figure 26. Matplotlib EDA Dashboard.

Methodology: The notebook combines nine Matplotlib subplots into a single analytical dashboard.

What the graph represents: This figure consolidates age trends, role counts, heatmaps, distributions, and symptom summaries in one page.

Significance: It demonstrates how exploratory visuals can be organized before formal BI dashboard development.

8.24 Seaborn EDA Dashboard



Figure 27. Seaborn EDA Dashboard.

Methodology: The notebook combines nine Seaborn subplots covering scatter, line, bar, count, box, violin, strip, swarm, and point plots.

What the graph represents: This dashboard focuses on lifestyle and mental health relationships with richer statistical plotting styles.

Significance: It bridges EDA and Power BI by showing which relationships deserve interactive dashboard treatment.

8.25 Correlation Matrix Heatmap

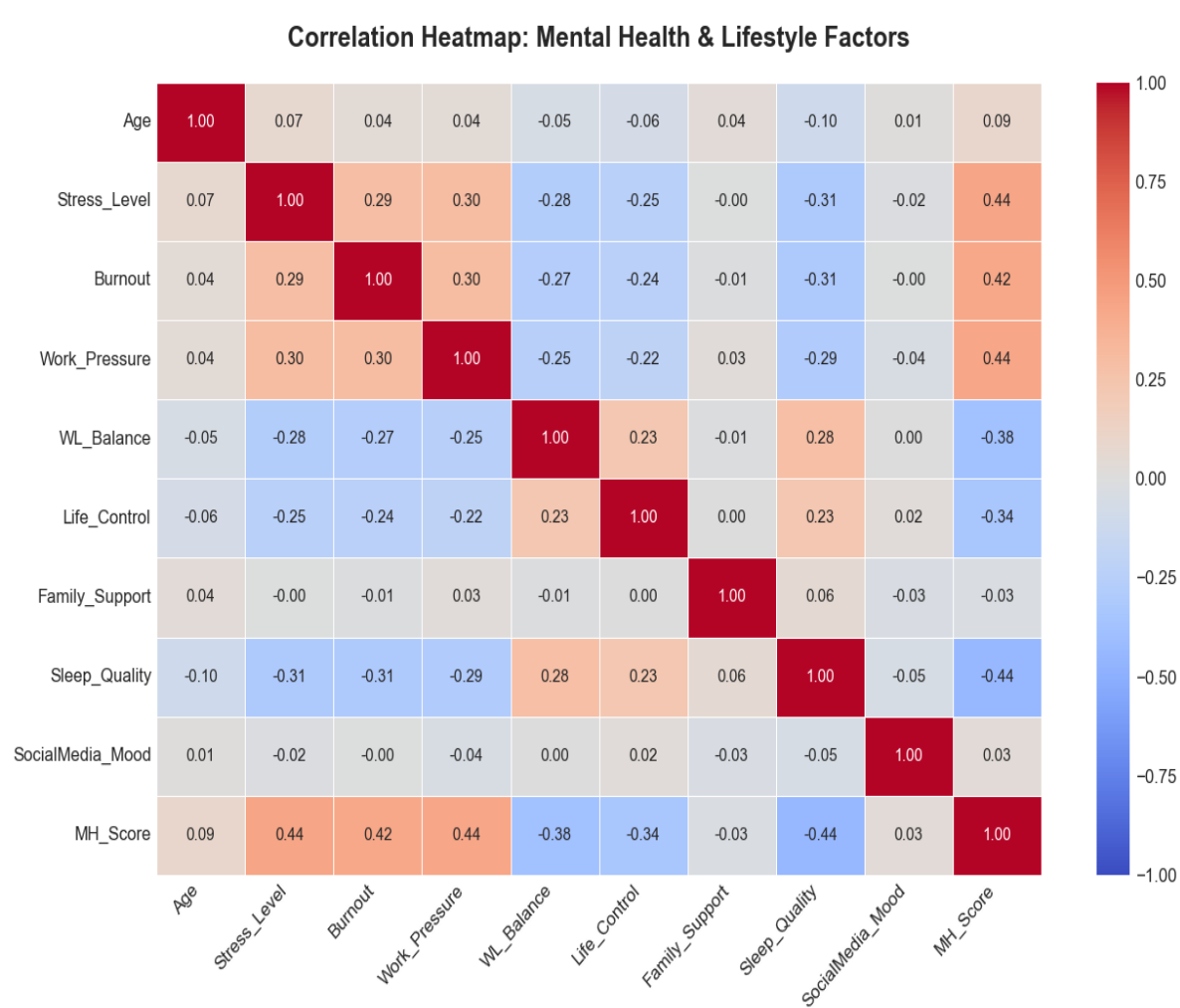


Figure 28. Correlation Matrix Heatmap.

Methodology: The notebook selects numeric columns and computes a Pearson correlation matrix.

What the graph represents: Important associations include Work_Pressure and Burnout (0.30), Stress_Level and Burnout (0.29), and Sleep_Quality with Stress_Level (-0.30).

Significance: Correlation analysis identifies the strongest numeric relationships while reminding that correlation does not prove causation.

9. Power BI Dashboard Analysis: Mental Health

The Mental Health dashboard is designed as a high-level monitoring page for respondent composition and mental-health indicators. It combines KPI cards, slicers, line charts, pie/donut visuals, role-based stress analysis, gender-segmented stress categories, symptom prevalence, and state navigation.

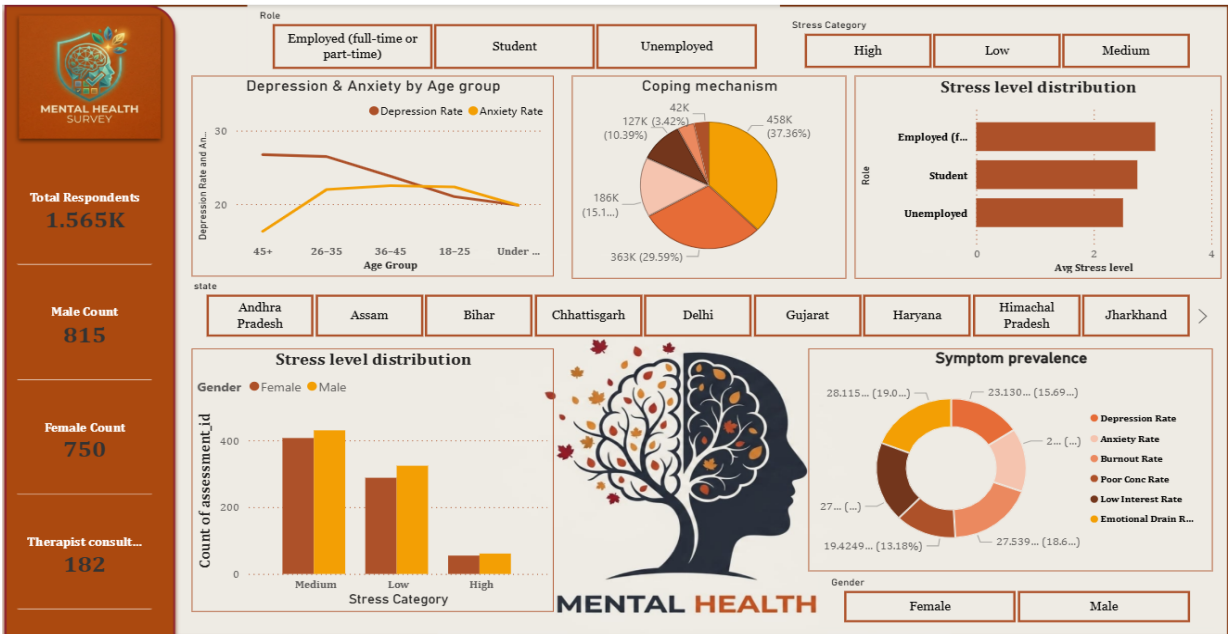


Figure 29. Power BI dashboard page: Mental Health.

Visual / Control	Displayed Value	Interpretation
Total Respondents	1.565K	Total number of survey respondents loaded into the dashboard model.
Male Count	815	Gender KPI enabling immediate comparison with female count.
Female Count	750	Completes the gender balance view and supports gender slicer analysis.
Therapist consultation	182	Respondents who reported consulting a therapist; highlights service-use gap.
Role slicer	Employed, Student, Unemployed	Filters all visuals by respondent role.
Stress Category slicer	High, Low, Medium	Filters visuals according to categorized stress score.

Visual / Control	Displayed Value	Interpretation
State slicer	Multiple Indian states	Enables geographic filtering and comparison.

The Depression & Anxiety by Age Group line chart compares depression and anxiety rates across age groups. The visual shows that psychological indicators do not remain flat across age; anxiety and depression vary across younger, middle, and older respondent groups. In dashboard use, selecting a role or stress category can reveal whether age-group patterns are driven by students, employed respondents, or unemployed respondents.

The coping mechanism pie chart divides respondents by coping approach. Entertainment is the largest visible segment, followed by talking to friends/family, exercise, not specified coping, substance use, and playing with pets. This visual is important because coping method quality affects intervention design; adaptive mechanisms such as exercise and talking to others are more protective than avoidant or harmful mechanisms.

The Stress level distribution by role chart shows average stress by respondent role. It aligns with the Python results in which employed respondents have the highest average stress and burnout. This chart is a management-facing visual because it quickly identifies which role group may require workload, schedule, or counseling support.

The Stress level distribution by gender and stress category compares male and female counts across low, medium, and high stress categories. The visual shows medium stress as the largest category for both genders, followed by low stress, with high stress being smaller but still important. This supports a targeted interpretation: most respondents are not in the extreme category, but a sizeable moderate-stress population may worsen if preventive supports are absent.

The symptom prevalence donut chart presents depression, anxiety, burnout, poor concentration, low interest, and emotional drain together. This creates a compact symptom-risk view. The EDA results show low interest and emotional drain as high average symptom dimensions, which should be referenced while reading the donut visual.

9. Power BI Dashboard Analysis: Mental Health

The Lifestyle and Wellbeing dashboard focuses on habits that may influence mental health: sleep quality, screen time, physical activity, work or study hours, state distribution, and categorical wellbeing filters. It is useful for identifying modifiable lifestyle patterns rather than only mental-health outcomes.

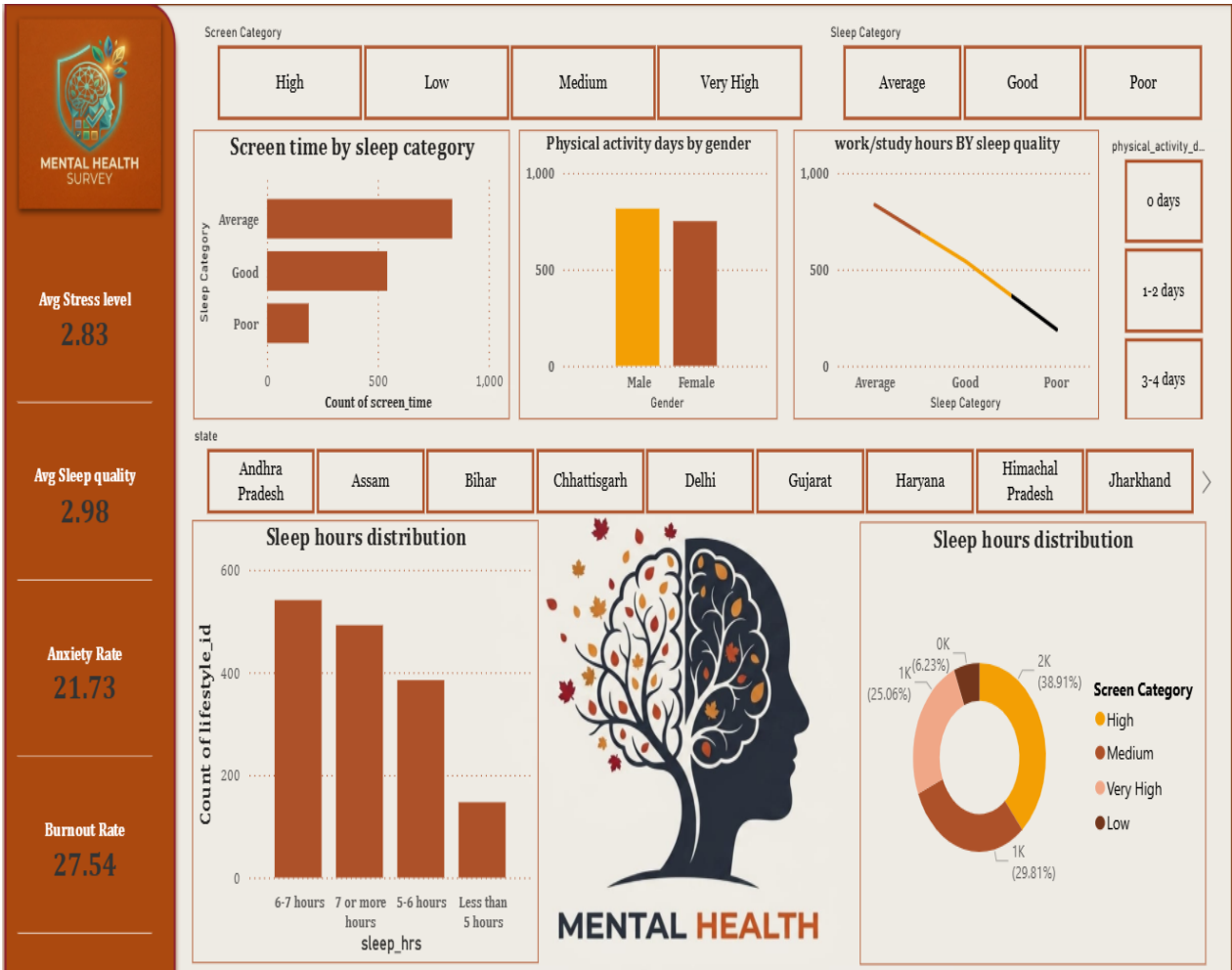


Figure 30. Power BI dashboard page: Lifestyle and Wellbeing Habits.

Visual / Control	Displayed Value	Interpretation
Avg Stress level	2.83	Average stress score across respondents.
Avg Sleep quality	2.98	Mean sleep quality score on the survey scale.
Anxiety Rate	21.73	Dashboard measure representing anxiety prevalence/rate.
Burnout Rate	27.54	Dashboard measure representing burnout prevalence/rate.
Screen Category slicer	High, Low, Medium, Very High	Filters charts by screen-time category.
Sleep Category slicer	Average, Good, Poor	Filters charts by sleep quality category.
Physical activity slicer	0 days, 1-2 days, 3-4 days	Filters lifestyle charts by activity frequency.
State slicer	State buttons	Filters page visuals geographically.

The Screen time by sleep category horizontal bar chart counts respondents by sleep category and screen time. Average sleep category has the largest visible count, followed by good and poor. This chart lets the analyst ask whether high or very high screen-time groups are overrepresented in poor sleep.

The Physical activity days by gender chart compares activity frequency counts between male and female respondents. It supports gender-aware lifestyle interpretation without reducing mental health outcomes to gender alone. In combination with the physical-activity slicer, it can identify whether inactivity is concentrated in a specific demographic segment.

The Work/study hours by sleep quality line chart shows that sleep quality worsens as the category moves from average to poor. This visual supports the EDA conclusion that lifestyle demands and rest quality interact with stress. A high workload with poor sleep is a practical risk pattern for burnout.

The Sleep hours distribution column chart shows sleep duration categories. The largest count is the 6-7 hours group, followed by 7 or more hours, then 5-6 hours, and finally less than 5 hours. This mirrors the workbook distribution and shows that most respondents are near moderate sleep duration, but a non-trivial group has insufficient sleep.

The donut chart for screen category breaks respondents into high, medium, very high, and low screen categories. The low category is visibly smallest, while high/medium/very high dominate. This reinforces screen exposure as a major lifestyle variable for student and employed groups.

11. Key Findings and Insights

- The project contains 1565 complete respondent records across three normalized tables and 33 merged analytical columns.
- Average stress level is 2.83, average burnout is 2.79, and average sleep quality is 2.98.
- 479 respondents (30.61%) reported affected mental health, while only 182 respondents reported therapist consultation.
- Employed respondents show the highest mean stress (3.05) and burnout (2.96).
- Work pressure and burnout have a strong positive correlation of 0.30.
- Sleep quality is inversely associated with stress; average stress falls from 3.38 at sleep quality 1 to 2.25 at sleep quality 5.
- Entertainment and talking to friends/family are the most common coping mechanisms, while therapist consultation remains relatively low.
- Power BI slicers enable role, state, gender, stress category, sleep category, screen category, and activity-based exploration.

12. Conclusion

The project successfully demonstrates an end-to-end analytics workflow from data storage to insight communication. The normalized database design separates respondent demographics, lifestyle habits, and mental health assessment measures while preserving relational integrity through primary and foreign keys. Python EDA validates data quality, creates derived variables, and reveals major patterns across stress, sleep, burnout, anxiety, coping, and support.

The Power BI dashboards convert these findings into interactive visual pages. The Mental Health dashboard focuses on respondent totals, role-based stress, symptom prevalence, coping mechanisms, and demographic filtering. The Lifestyle and Wellbeing dashboard focuses on sleep, screen time, activity, work/study hours, and wellbeing KPIs. Together, the project shows how technical analytics can support awareness, early identification of stress patterns, and data-driven wellbeing interventions.

13. References

- World Health Organization. Mental health and wellbeing public health resources.
- PostgreSQL Global Development Group. PostgreSQL Documentation: CREATE TABLE, constraints, and foreign keys.
- Pandas Development Team. Pandas documentation for dataframe operations, joins, descriptive statistics, and missing-value analysis.
- Matplotlib and Seaborn documentation for statistical visualization methods.
- Microsoft Power BI documentation for dashboard visuals, slicers, filters, cards, and measures.
- Supplied project files: Mental_Health_Survey_.xlsx, Mental Health-Survey_EDA-Using-Python-new.ipynb, Full_Name.pdf, ER diagram.jpeg, and Power BI dashboard screenshots.

14. Appendix

Appendix A: Complete Merged Column List

No.	Column Name
1	Respondent_ID
2	Age
3	Gender
4	State
5	Role
6	Lifestyle_ID
7	Work_Study_Hrs
8	Sleep_Hrs
9	Physical_Activity_Days
10	Screen_Time
11	Outings_Freq
12	Sleep_Quality
13	SocialMedia_Mood
14	Regular_Breaks
15	Has_Coping_Strategy
16	Assessment_ID
17	Low_Interest
18	Depressed
19	Sleep_Trouble
20	Low_Energy
21	Poor_Concentration
22	Anxiety
23	Emotional_Drain
24	Stress_Level
25	Burnout
26	Work_Pressure
27	WL_Balance

No.	Column Name
28	Life_Control
29	Family_Support
30	Coping_Mechanism
31	Therapist_Consult
32	Social_Support
33	MH_Affected

Appendix B: Category Counts

Work_Study_Hrs

Category	Count
6-8 hours	612
4-6 hours	464
More than 8 hours	320
Less than 4 hours	169

Sleep_Hrs

Category	Count
6-7 hours	541
7 or more hours	492
5-6 hours	385
Less than 5 hours	147

Physical_Activity_Days

Category	Count
1-2 days	586
3-4 days	437
0 days	343
5+ days	199

Screen_Time

Category	Count
4-6 hrs	617
2-4 hrs	466
6+ hrs	387
Less than 2 hrs	95

Outings_Freq

Category	Count
1-2x per week	616
Rarely	458
3-5x per week	296
Almost daily	195

Regular_Breaks

Category	Count
Yes	870
No	695

Has_Coping_Strategy

Category	Count
Yes	1158
No	407

Coping_Mechanism

Category	Count
Entertainment (music, movies, gaming)	598
Talking to friends/family	466
Exercise	221
Not Specified	159
Substance use	63
Playing with pets	58

Appendix C: Dashboard Interaction Logic

Power BI slicers act as report-level or page-level filters depending on configuration. In this project, role, gender, state, stress category, sleep category, screen category, and physical-activity controls allow users to isolate subgroups. When a slicer selection is made, cards, charts, and donut visuals recalculate under the selected filter context.